



Survey paper

Blockchain and federated learning-based intrusion detection approaches for edge-enabled industrial IoT networks: a survey

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ABSTRACT

The industrial internet of things (IIoT) is an evolutionary extension of the traditional Internet of Things (IoT) into processes and machines for applications in the industrial sector. The IIoT systems generate a large amount of private and sensitive data i.e., stored and processed somewhere on the cloud-edge continuum. The IIoT devices, and the IIoT networks are subject to security mechanisms such as intelligent Intrusion Detection and Prevention Systems (IDS/IPS) systems, that can detect and respond unseen malicious network attacks. The adoption of centralized machine learning methods for IDS has become impractical due to the high computational cost and privacy concerns associated with storing large amounts of data on a single server along the cloud-edge continuum. The combination of federated learning and Blockchain has emerged as a promising advancement in addressing the challenge. Federated learning distributes learning to individual IIoT devices without compromising data privacy, while Blockchain enhances privacy and security. Many academic and industrial efforts outline IDS mechanisms using machine learning, deep learning, federated learning, and Blockchain technologies. The utilization of federated learning-based IDS has become increasingly popular and is now being applied to various tasks including IDS/IPS systems. However, existing intrusion detection systems (IDSs) survey are limited to the scope of classical machine learning and deep learning. To address this limitation, we analyze the IIoT literature that integrates Blockchain and federated learning to enhance IDSs and improve its threat detection capabilities. This survey explores the role of Blockchain and federated learning in addressing security and privacy issues, particularly those associated with IDS/IPS in IIoT networks. Insights on the possibilities of machine learning, federated learning, and Blockchain in supporting IDS to monitor IIoT network traffic for anomaly detection are discussed in detail through state of the art. Furthermore, we provide a set of recommendation based on our literature for the effective implementation of a Blockchain and federated learning-based network intrusion detection system. Finally, we summarize the study and highlight challenges as future research directions for Blockchain and Federated Learning-based technologies for cybersecurity and intrusion detection in IIoT.

1. Introduction

Industry 4.0 has emerged through the seamless integration of the internet of things (IoT) into industrial processes and infrastructure. This integration has enabled the transformation and advancement of modern industries. This new era involves millions of intelligent industrial devices interacting with or without human intervention. IoT and Industrial IoT (IIoT) networks are leveraged to connect various electro-mechanical devices, such as production equipment, instruments, sensors, and actuators, present in real-world industrial setups [118]. These connected

devices enable real-time decision-making powered by the extensive computing capabilities of the cloud-edge continuum. The cloud-edge continuum refers to the distribution and coordination of computing resources and data processing between cloud computing infrastructure and edge devices, such as sensors, gateways, and edge servers. In this continuum, computing resources are distributed across a spectrum, ranging from centralized cloud data centers to the edge of the IIoT network. This decentralized approach allows for efficient data collection, processing, and storage of industrial big data generated by numerous sensors. At the edge of the IIoT network, edge devices play a

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crucial role in executing initial data filtering, analysis, and preprocessing. This essential process effectively reduces the volume of data that must be transmitted to the cloud, leading to reduced latency and bandwidth requirements. As a result, the IIoT system becomes more efficient and responsive. Consequently, real-time or near-real-time decision-making becomes achievable at the edge, while more complex analytics, long-term storage, and resource-intensive computations are offloaded to the cloud. The cloud-edge continuum thus facilitates the implementation of advanced technologies within Industry 4.0 and ensures efficient management of Industry 4.0 processes.

The Industrial Internet of Things (IIoT) significantly enhances industry productivity by integrating digital devices, services, Supervisory Control and Data Acquisition (SCADA), and physical systems. This seamless integration of technology and physical infrastructure has the potential to revolutionize various industries, bringing forth increased efficiency and advanced capabilities. The IIoT framework brings numerous advantages across various sectors, including automotive, transportation, healthcare, and more [29,34]. In a comprehensive study, Piccialli et al. [123] thoroughly explored the impact of IIoT on various industrial sectors, highlighting its potential to enhance services and products. The authors delve into specific areas where IIoT benefits will drive organizational transformation and the emergence of novel business models facilitated by IIoT solutions. Moreover, IIoT, empowered by the cloud-edge continuum, has become the focus of new research trends for industrial applications, aiming to improve the efficiency of industrial processes [188]. The ongoing IIoT digital revolution leads to profound systemic changes from traditional to digital industrial operations. This digitalization and cutting-edge technology using IIoT enable management executives to monitor industrial processes and visualize the hidden trends embedded in the massive amounts of data generated by the IIoT physical layer infrastructure. The goal of the IIoT is to enable intelligent manufacturing, facilitating the establishment of smart factories with effective communication between business partners and customers. This entails leveraging interconnected devices and systems to create a highly efficient and collaborative industrial ecosystem [15]. IIoT integrates data, services, and people for intelligent operations across various industries, such as smart electricity, smart cities, healthcare, the automation industry, agriculture, logistics, and transportation [78]. However, the IIoT network requires additional specific security measures as it is vulnerable to attacks, and the compromise of even a single device can have far-reaching effects due to the interconnected nature of the system and the high value of the data it contains. The authors in [116] have projected that the absence of effective intrusion detection systems (IDS) in the near future could result in cyber threats within the IIoT, incurring costs of up to \$90 trillion by 2030. The primary peril within the IIoT lies in the prevalence of malware, where attackers exploit vulnerabilities in computers to carry out malicious activities such as denial of service (DoS), decentralized DoS (DDoS), and progressive determined risk (PDR) attacks. To detect and minimize intrusions, several security measures can be employed. An intrusion detection system (IDS) is a cybersecurity monitoring system designed to prevent unauthorised access to computer networks. An intrusion occurs when there is unauthorised access to a system, compromising its confidentiality, integrity, and availability (collectively referred to as CIA). In network security, intrusion detection plays a vital role in identifying anomalies in network activity, ensuring the protection of the CIA triad. As the IIoT system rapidly evolves, its widespread deployments present a plethora of challenges, such as ensuring confidentiality, enhancing data accountability, and ensuring availability [105].

Within the IIoT ecosystem, the intrusion detection system (IDS) plays a crucial role in safeguarding sensitive information. Despite the presence of security mechanisms like IDS and end-point protection, it is important to note that they may not be fully effective against all potential attacks [6]. Several articles have presented IDS mechanisms for IIoT systems, employing attack signatures stored in databases to identify attackers [15,54]. To establish a comprehensive and robust security posture for

IIoT scenarios, a diverse range of security countermeasures and technologies can be utilized. These include reactive, proactive, preventative, remedial, forensic, and intelligence approaches. Such a multi-layered and versatile approach is essential to effectively protect IIoT systems from various threats and ensure their smooth operation and resilience against potential cyberattacks [162,7]. Despite such efforts, IIoT remains susceptible to various threats, including distributed denial of service (DDoS), Man-in-the-Middle, Ransomware, and others, posing challenges in distinguishing between legitimate and malicious activities [152,184]. In order to address these challenges, it is crucial to implement an adaptive IDS capable of identifying potential cyber-attacks with minimal false positives and no disruptions to mission-critical services. The research community is actively involved in the development of reliable and robust intrusion detection tools tailored for IIoT scenarios to enhance network security in the IIoT domain [107,166].

In recent times, there has been a notable surge of interest in addressing cybersecurity concerns within IIoT ecosystems, particularly focusing on intrusion detection systems (IDSs) based on federated learning (FL) [45]. FL endeavors to create a distributed machine-learning approach similar to incremental learning but without the need for explicitly sharing data samples. It emphasizes data heterogeneity, availability, and preserving privacy while reducing model bias, making it a promising machine learning paradigm for IIoT [66]. Additionally, FL provides a distributed framework that ensures privacy protection and promotes collaboration among numerous participants, enabling iterative training of machine learning models tailored to IIoT applications. However, FL-based IDS algorithms face several challenges in real-world applications. Firstly, finding a device with sufficient capacity and resources to serve as a central server can be challenging in certain IIoT scenarios. Secondly, FL is vulnerable to central server failures, as any malfunction or disconnection can disrupt collaborative training. Thirdly, scalability issues arise, resulting in a communication bottleneck when the number of participating devices increases.

Blockchain technology most effectively conforms to an IIoT cybersecurity framework through the use of action research, which may be interpreted as the collection of quantitative, qualitative, or mixed data. From the security perspective, it is vital to incorporate Blockchain technology for the IIoT network and increase security. Blockchain has the potential to address these needs and play a vital role by providing verifiable and secure information exchange and storage options [168]. By combining Blockchain with FL, intrusion detection can be significantly improved, providing distinct advantages in terms of security and privacy protection [83]. Furthermore, Alruwaili et al. [194] highlighted the fundamental role of intrusion detection in IIoT systems. The authors in [194] presented a wide array of protocols, algorithms, and mechanisms specifically designed for intrusion detection in the context of IIoT. Moreover, they proposed the integration of machine learning (ML) and Blockchain technology as an effective means to mitigate security threats within the IIoT domain. This innovative approach holds great promise in bolstering the security and resilience of IIoT systems against potential cyber threats. Common needs in IIoT applications, such as protecting the confidentiality of sensitive information and fostering mutual trust among parties involved in different stages of the supply chain, underscore the significance of leveraging these technologies. Existing state-of-the-art technology lacks the essential integration of FL and Blockchain for IIDS in IIoT networks, compromising security and privacy. This integration is crucial for robust cybersecurity solutions. Combining Blockchain and FL for IDSs in IIoT networks offers enhanced security, privacy, real-time threat response, resilience, and compliance capabilities. Given the susceptibility of IIoT systems to cyber threats and attacks with potentially severe consequences for critical infrastructure and industrial operations, integrating intrusion detection mechanisms becomes imperative to bolster security and safeguard IIoT devices, networks, and data. With its decentralized and immutable ledger, Blockchain technology enhances trust and transparency in IIoT systems, while FL enables collaborative model training without sharing sensitive

data, preserving privacy and confidentiality. IDSs can analyze network traffic, device logs, and system behavior to detect potential threats or abnormal activities, enabling organizations to promptly detect and respond to emerging threats through continuous monitoring of IIoT networks and devices. Additionally, FL facilitates the rapid distribution of updated intrusion detection models across IIoT devices, enabling real-time threat mitigation without relying on centralized systems.

In our study, we followed a systematic approach presented in Fig. 1. to gain valuable insights into prevailing trends and methodologies from existing IIoT literature on IDS/IPS. Employing a structured approach, we thoroughly explored and analyzed the literature, providing us with a holistic understanding of the current state of research, identifying knowledge gaps, and proposing potential avenues for future investigations. Our primary objective is to offer a comprehensive overview of the existing research landscape, identify areas requiring further exploration, and propose promising directions for future studies. This article presents a survey of research articles focused on IDS for the IIoT setup that leverages Blockchain and distributed machine learning technologies. The key contributions of this survey are as follows:

- We provide a detailed analysis of survey articles related to network intrusion detection for networked systems. It examines existing literature on intrusion detection techniques in the context of IIoT and identifies trends and advancements in this field.
- We present an in-depth and extensive review of the diverse array of attacks directed toward IIoT systems. This article delves into the intricacies and nuances of these attacks, providing a fluent and provides thorough exploration of IDS/IPS techniques in IIoT.
- We conduct a comprehensive literature review of network intrusion detection methods tailored explicitly for IIoT environments. Based on our understand from the literature we provide a set of guidelines on the utilization of Blockchain and federated learning techniques to enhance the security and efficiency of intrusion detection systems for IIoT.
- This survey compares the various literature reviews based on different parameters. These parameters may include the types of intrusion detection methods, the use of Blockchain and federated

learning, performance metrics, and application domains, among others.

- We debate and engage in discussion about open issues and propose potential directions for future research in intrusion detection for IIoT. These discussions highlight areas requiring further exploration and improvement, fostering advancements in the IIoT scenario.

The remainder of the survey paper is structured as follows: Section II presents an overview of the related survey papers. Section III presents background on IDS in IIoT networks, considering Blockchain as an auxiliary cybersecurity technology and Federated learning privacy-preserving distributed machine learning approach that fits the distributed IIoT networks. Section IV presents a discussion on the possible threat to IIoT networks. Section V analyzes the current state-of-the-art Blockchain and FL-based IDS techniques in IIoT Networks and compares and contrasts them. Section VI presents the research challenges and open issues for future exploration and improving IDS in IIoT networks. Finally, section VI concludes the manuscript.

2. Related survey papers

IDS systems have been widely studied in the context of computer networks. This is clear from the number of survey studies published on IDS in computer networks. Table I. summarizes and discusses these survey papers.

Many research efforts have been devoted to surveys and literature covering machine learning (ML), FL, and Blockchain-based solutions for IIoT networks. Table 1. presents the list of survey articles under the scope of IoT/IIoT applications. According to a recent study by Zhao et al. [187], Blockchain-based technology has several benefits from an IIoT point of view. Additionally, they briefly described the IIoT networks powered by Blockchain and highlighted several fundamental problems with the IIoT architecture. The authors in [144] addressed the vulnerabilities and attacks targeting IoT and IIoT networks. To tackle the security issues in IIoT, they conducted an in-depth analysis of a privacy solution based on Blockchain technology. In their research the authors in [106] proposed a comprehensive examination of Ethereum-based Blockchain for IoT networks. The authors in [134] examined the

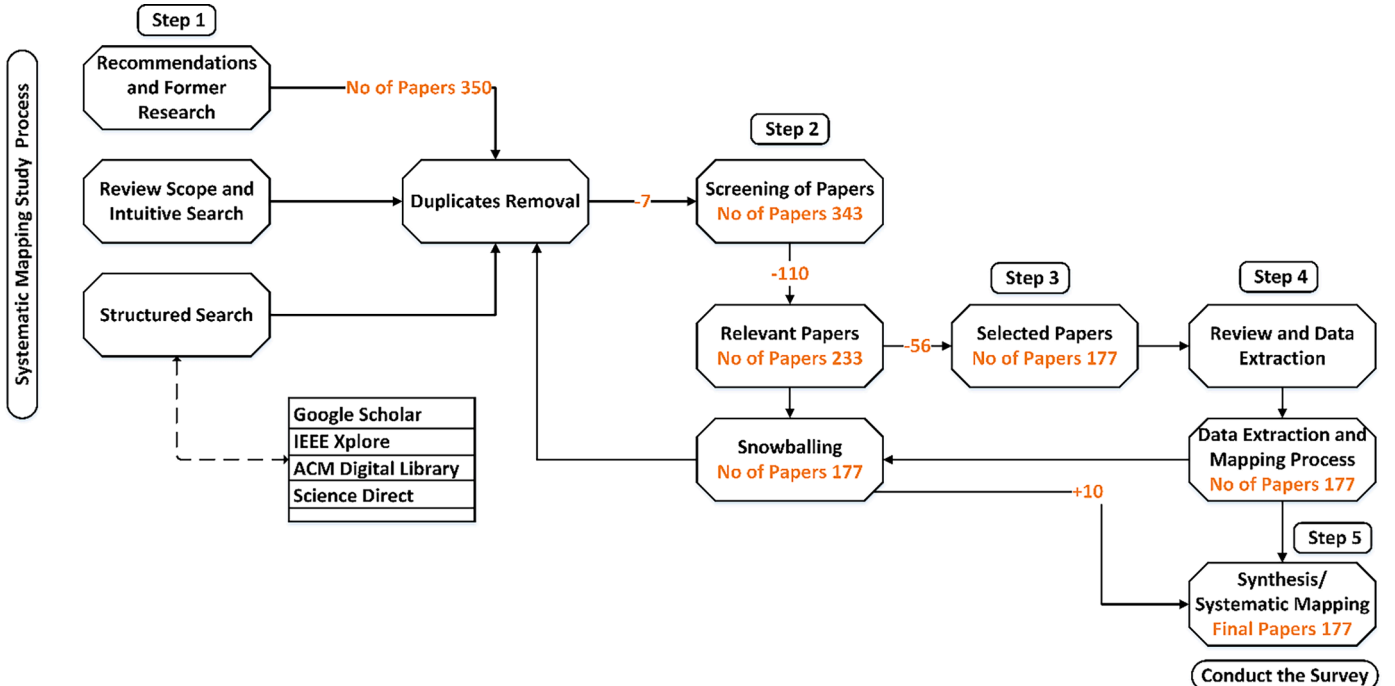


Fig. 1. Systematic mapping of the study.

Table 1
Related Surveys on IDS in IIoT Networks.

Reference	Year	Contribution	IDS method	Blockchain focused	FL focused	ML focused
[62]	2022	This study focuses on a comprehensive layered architecture in an advanced research framework. It provides an overview of cyber-physical interaction, identity resolution systems, artificial intelligence, 5 G/B5G, 6 G, and edge computing. The architecture includes protocol, data, consensus, and control layers, aiming to establish a robust and efficient system.	N/A	✓	×	×
[44]	2022	The authors propose Edge-IIoTset, a novel dataset for assessing cybersecurity in IoT and IIoT. It incorporates centralized, federated learning, and machine learning-based IDSs. Generated using a custom-built IoT/IIoT testbed, the dataset includes various attack types using diverse IoT devices.	Hybrid	×	✓	✓
[90]	2021	This review explores technologies for passively collecting wireless signal patterns and network traffic traces to identify IoT devices. It discusses advancements in machine learning and deep learning techniques, including intelligent, anomaly detection, and deep unsupervised learning, focusing on emerging developments.	NIDS	×	×	✓
[73]	2021	Presented a review of recent works to improve intrusion techniques, deployment strategy, the taxonomy of attacks, and IDS strategy for IoT security.	Anomaly	×	×	✓
[167]	2021	This article combines machine learning with Blockchain to identify and mitigate real-time IIoT network attacks. The suggested state-of-the-art focuses on standalone security solutions and does not consider a cohesive technique for monitoring IIoT networks and finding edge intruders.	Traditional IDS	✓	×	✓
[161]	2021	A detailed analysis discusses IIoT security and four-layer security architecture, examines deployed countermeasures based on CIA+ security criteria, and gives insight into today's industrial countermeasures. The proposed research development and the persistent challenges of the IIoT ecosystem are also discussed.	N/A	×	×	×
[104]	2021	Security concerns and issues are notably widespread across various layers of the IoT network architecture, including the perception layer, network layer, support layer, application layer, and data layer. Particular attention is given to topics like anomaly detection, various IDS techniques, and the growing threat of DDoS attacks.	Hybrid	×	×	✓
[144]	2020	The study categorizes and maps attacks, responses, and vulnerabilities in IoT/IIoT systems. It further compares the advantages of integrating Blockchain with IoT/IIoT applications and explores Blockchain-based IoT solutions. Additionally, the practical aspects of incorporating Blockchain with IoT are thoroughly discussed.	Centralized	✓	×	×
[106]	2020	Outlined how to solve security and privacy concerns with IoT and blockchain technology integration. Authenticated IoT devices using Ethereum and discussed the security analysis.	N/A	✓	×	×
[21]	2020	The most cutting-edge uses of Blockchain technology in several Industry 4.0 and intelligent applications are inspected. However, it also detailed the advantages and disadvantages of currently available Blockchain-based security solutions.	N/A	✓	×	×
[49]	2020	The authors provide a comprehensive overview of IIoT to address security vulnerabilities. They propose a taxonomy for the underlying infrastructure, offer security guidelines, and analyze appropriate security protocols across different tiers to achieve secure connectivity in the IIoT.	N/A	×	×	×
[94]	2020	In this article, the focus is on IIoT in smart cities, with an emphasis on the security challenges it poses. The authors employ deep learning techniques, such as reinforcement learning, recurrent neural networks, and convolutional neural networks, to enhance IoT security. The paper delves into security approaches and future trends, aiming to improve IIoT security through the implementation of DL techniques.	N/A	×	×	✓
[162]	2020	The systematic review on IIoT security mainly focuses on the new paradigm of Fog computing to improve the security requirements based on IIoT.	N/A	×	×	×
[145]	2020	A comprehensive approach is recommended to safeguard the growing IIoT domain during the transition to Industry 4.0. It considers security risks from consumer IoT and acknowledges distinctions between consumer and industrial sectors, addressing imminent security challenges in protecting IIoT systems.	Centralized	×	×	×
[33]	2019	The research explored the integration of Blockchain technology with IoT, focusing on industrial applications called Blockchain of Things (BcoT). It introduced the convergence of Blockchain and proposed an IoT architecture for BcoT.	N/A	✓	×	×
[187]	2019	The study focuses on the integration of Blockchain and IIoT from an industrial perspective. It presents a Blockchain-enabled IIoT architecture that utilizes IoT platforms and incorporates the use of Blockchain technology.	N/A	✓	×	×
[14]	2018	The study discusses challenges in decentralizing IoT and explores the incorporation of Blockchain-based solutions. It addresses privacy, trust, security, identity management, data management, and monetization issues in IoT, specifically focusing on Blockchain consensus algorithms.	N/A	✓	×	×
[43]	2018	The paper describes various IoT application areas and threat models, including identification, manipulation, and cryptanalysis. It also highlights recent advancements in Blockchain technologies that aim to enhance security and privacy for users.	Traditional IDS	✓	×	×
[137]	2018	Survey on Blockchain technology, identification and analysis of several approaches to integrating IoT with Blockchain, and evaluating the performance of various Blockchains in IoT devices.	N/A	✓	×	×
[25]	2018	This review paper combines IIoT and analyses the linkages of relevant IoT taxonomies to concepts such as cyber-physical systems and Industry 4.0 security risks and vulnerabilities.	N/A	×	×	×

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Table 1 (continued)

Reference	Year	Contribution	IDS method	Blockchain focused	FL focused	ML focused
[155]	2018	The study shed light on important concepts and challenges in IoT/IIoT, particularly concerning energy efficiency, real-time performance, cohabitation, and interoperability. It also addresses security and privacy concerns in the current trend of automation and data interchange in Industry 4.0.	N/A	×	×	×
[111]	2018	The proposed paper outlines an innovative security-by-design approach for IIoT, encompassing the design/modeling and runtime/simulation stages. This approach facilitates the analysis of security requirements, identification of potential attack vectors, and their integration to effectively minimize security breaches.	N/A	×	×	×
This Survey	2022	We investigate the IDS/IPS security solutions for IIoT systems that utilize Blockchain, Federated Learning for the detection of malicious attacks on IIoT networks. Furthermore, we provide a discussion on open issues and future research directions for intrusion detection in the IIoT scenario.	All	✓	✓	✓

significance of Blockchain technology in the context of IIoT applications. Moreover, methods are also presented in their Blockchain model for the IIoT networks. The researcher in [21] presented a comprehensive taxonomy of Blockchain-enabled IIoT networks along with a thorough analysis of current centralized systems. Blockchain's potential IIoT applications were also highlighted in this research. The study conducted by Monrat et al. [109] presented a comparative analysis of various Blockchain designs for the IoT. They considered a wide range of consensus mechanisms and techniques to evaluate their applicability and effectiveness in IoT scenarios. The research focused on evaluating the effectiveness and suitability of various consensus techniques within the context of IoT applications and examined how different Blockchain designs perform. Furthermore, this paper also discusses the future scope and potential of Blockchain applications for IoT. The authors in [33] highlighted a brief introduction of Blockchain protocols to identify and address IoT network security challenges. The authors in [14] delved into potential research challenges and future initiatives related to Blockchain-based IoT systems. Additionally, they provided a brief overview of the key characteristics associated with these systems. In their proposal the authors in [43] presented a thorough evaluation of the current state of Blockchain and IoT technologies. The authors also demonstrated the usefulness of Blockchain technology in intelligent transportation and energy management, and they analyzed the many forms of attack and threat models in IoT networks. Reyna et al. [137] examined the critical issues plaguing IoT networks and explained in depth how deploying Blockchain platforms in the cloud-edge continuum might boost the efficiency and effectiveness of the networks. The work proposed in [62] offers a thorough and comprehensive assessment, analyzing the use and integration of Blockchain in IIoT contexts, while also providing a review of the significant concerns confronting IIoT. However, the authors in [73] provided a thorough analysis of the techniques, deployment, validation strategies, datasets, and technologies used in IoT intrusion detection systems, along with their benefits and drawbacks. In this work, they also shed light on IoT security concerns and looked at current models for IoT IDS performance enhancement. The authors in [39] introduced an enhanced IDS that leverages gradient boosting (GB) and decision tree (DT) algorithms. They implemented these algorithms using the open-source Catboost platform, specifically targeting IoT security. The primary objectives of this approach are to improve classification performance, increase accuracy and precision of the IDS, and reduce detection time. To handle the large volume of data efficiently, the system incorporates multi-GPU implementation support, reducing both processing time and detection time.

The authors in [90] present a complete review of the technologies now available for passively collecting wireless signal patterns and network traffic traces in order to detect and identify IoT devices. In addition, Machine Learning and deep learning for IoT device identification have discussed many important emerging developments,

including incremental learning (IL), abnormality detection, and deep unsupervised learning with an endowment. As a means of addressing the unique security challenges posed by IIoT, Tan et al. [161] provides a thorough assessment of the relevant literature and focuses on how IIoT security architecture differs from existing systems. Moreover, the IIoT architecture has each layer subjected to a comprehensive end-to-end security review. The devised model used in this survey covers the security needs by using the CIA model, highlighting recent industry countermeasures and flaws, discussing ongoing security difficulties, and discussing future security challenges, also includes using a Bottom-up method to create a standardized architecture for protecting IIoT devices. The authors in [104] conducted a review, analyzed the intrusion detection system, and evaluated several models. Additionally, researchers have studied the categorization of IDSs and several anomaly detection approaches, IDS models are constructed using datasets, machine learning, and deep learning techniques to preprocess data and effectively identify malware. The author in [58] analyzed an IDS-SioEL, a new intrusion detection framework for IoT-based smart environments that utilizes Ensemble Learning. The framework was evaluated using GPU processing on three datasets (IoT-23, BoT-IoT, and Edge-IIoT). The researchers utilized an efficient anomaly detection model, employing AdaBoost, and integrated it with a range of feature selection techniques, such as Boruta. The authors in [25] presented an analysis methodology for IIoT devices, which offers a useful classification schema for security-related issues in IIoT networks. Tange et al. [162] highlighted the current state of knowledge about security for the IIoT and constructed methodological search queries on several literature repositories. The study also highlighted taking a fog computing viewpoint and outlining what constraints and problems need to be solved to achieve broad fog computing adoption in the industrial environment. Using different techniques the authors in [49] provided an outline of IIoT's security and privacy measures concerning understanding where different security protocols are at different layers, emphasizing and identifying to solve existing security and privacy challenges. At the same time, the study conducted by Sisinni et al. [155] gave an overview of the architecture and protocols of emerging IIoT solutions provided in this research with a view to standardization. The study concentrated on the energy efficiency, real-time performance, cohabitation, interoperability, security, availability, and privacy; the research looked ahead to the future challenges that IIoT would face. In [111], authors proposed a secure simulation method for developing and designing a secure IIoT environment. The proposed approach by Serror et al. [145], the researchers outlines the distinct security goals and difficulties of the IIoT, which primarily stem from productivity and safety concerns in contrast to consumer deployments. Some future directions are also provided for the automatic implementation of intrusion detection, such as security mechanisms for resource-constrained devices in IIoT network time-based industrial processes and secure configuration of long-lived

industrial components. The authors in [94] outlined the notion of IIoT application employed for smart cities and smart homes; however, in contrast, they also presented the existing issues in IIoT security. The advantages and disadvantages of security-related systems for IIoT in smart cities are discussed taking into account the integration of machine learning and deep learning methods. Moreover, this article provides explanations for various deep learning methods, including Deep Reinforcement Learning, Recurrent Neural Networks, and Convolutional Neural Networks. The authors in [44] introduced an extensive cyber security dataset specifically designed for ML and intrusion detection. This dataset is tailored to support federated learning in IoT and IIoT environments. The survey article [167] integrated a Blockchain and ML-based integral technique for IoT devices to identify the threat and activate secure information transfer mechanisms to allow to modify the computational competencies of IIoT networks. The proposed study also integrated to restraint intrusion from the edge side.

3. Background

3.1. Intrusion detection systems

An Intrusion Detection System (IDS) is a way of detecting intrusions for protecting and monitoring abnormal data and events happening within a network or on a local computer from unauthorised access, which breaches security. The authors in [108] introduced a cutting-edge network intrusion detection model explicitly tailored to address the unique challenges of IoT environments. The model effectively utilizes the K-nearest neighbors (K-NN) classifier and incorporates feature selection techniques to enhance its accuracy and efficiency. Furthermore, the main component is to enhance the accuracy (ACC) and detection rate (DR.) of the IDSs. The researchers employ a comprehensive approach to feature selection, utilizing principal component analysis (PCA), univariate statistical tests, and genetic algorithms (GA) to enhance data quality. The proposed model's evaluation is conducted on the Bot-IoT dataset, demonstrating its effectiveness and robustness. IDS provides comprehensive solutions for mitigating and detecting malicious activity by thwarting security breaches. We may categorize intrusion detection systems as either network-based, host-based, or hybrid. The Hybrid-IDS monitors a single computer system for any indications of malicious behavior. Simultaneously, the Network-IDS analyses the traffic on the network in search of any suspicious payloads as intrusion detection systems protect Fig. 2. Shows network hosts and external networks for monitoring any suspicious behavior by the intruder. Furthermore, IDS is classified into two distinct categories based on detection techniques: signature-based and anomaly-based [103].

Developing a holistic environment for IDS is an exceedingly challenging task for IIoT using ML-based detection methods. IoT edge devices have limited resources and cannot process large volume of data and learn ML models. However, ensuring data security and privacy for a vast amount of heterogeneous IIoT data poses significant challenge that needs to be detected from potential security threats with a few false alerts before they become a catastrophe. Fig. 3 shows, the main steps in

IIoT ecosystem for IDS implementations. It is necessary to adopt multi-level protocols for heterogeneous communication infrastructure. A significant trust in defending IIoT Networks from Cyber Attacks is complicated by a high trust boundary and limited upgrade opportunities [55]. With the goal of addressing ML-based IDS for associated IoT and IIoT technologies [32,52,139], including wireless communication networks [126], and cyber-physical systems, specific review articles have been proposed.

Moreover, the architecture comprises three distinct parts of an IIoT system: the network layer, the physical layer, and the application layer [189]. The physical or perception layer consists of IoT sensors and devices. The network layer is responsible for transmitting sensing data from the perception layer's IoT devices to the application layer through Ethernet or Wi-Fi communication. Many applications on the application layer utilize servers and databases [120]. However, ML/DL-based approach for an IDS network architecture is shown in Fig. 4, where IIoT is vulnerable to attacks like DDoS, Man-in-the-Middle, Ransomware, etc. In order to prevent unauthorised access, a firewall is set up in a security system against suspicious activity. Nevertheless, this approach may not be the most effective way to secure a large system like IIoT. Additionally, Fig. 5 shows three-tier architectures of the IIoT environment.

Hence, it is essential to offer a recent and comprehensive systematic overview of IDS-based cybersecurity for IIoT, emphasizing the challenges and considerations encountered by researchers when designing a suitable security system for the IoT. Additionally, providing an up-to-date systematic evaluation of cyber security for IIoT is of utmost importance.

3.2. Blockchain

The inception of Blockchain was originally as a ledger for the Bitcoin network. It operates as a decentralized and append-only register, comprising a chain of blocks. The primary objective of Blockchain is to uphold an unchangeable and consistent record of peer-to-peer transactions. Each block within the Blockchain contains transaction records, collectively ensuring the system's integrity and security [171,117,66,65]. In addition, Blockchain offers limitless potential for IIoT applications due to its exceptional characteristics. Blockchain-based IIoT systems leverage encryption and authentication procedures to ensure user data privacy and protect industrial applications. Significant progress has been made in studies on Blockchain-based privacy protection since 2020. The transparent nature of Blockchain and the ability to trace and verify IDS activities can aid in regulatory compliance and demonstrate due diligence. Accessing a block and its associated data records in a manner that is both sequentially and cryptographically secure is the primary benefit offered by a Blockchain.

Furthermore, Blockchains are frequently employed as a public record of transactions, although they are typically shared and synced via a distributed system. Each Blockchain member can see, discard, or check the record depending on the protocol. It also begins the consensus mechanism and adds the new authenticated block to the Blockchain [174]. Blockchains have a lot of potential to enhance current IIoT platforms' performance. Big data, IoT, artificial intelligence (AI), and smart robotics are transforming industrial processes [72]. Blockchain-enabled IIoT holds the potential to significantly benefit a diverse array of industries, including smart manufacturing, transportation, healthcare, energy, and various others. The integration of Blockchain with IIoT platforms enhances the overall security of the entire IIoT system. Fig. 6 shows the smart industrial environment architecture.

Blockchain is widely assumed to be a novel distributed computing and storage paradigm that merges multiple current technologies [183]. Using cryptographic proofs, participants communicate transactional data over a vast network of untrusted nodes. Additionally, Blockchains offer a computational infrastructure capable of running computer

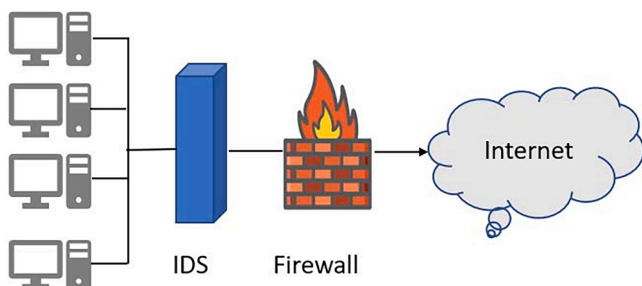


Fig. 2. A network intrusion detection system.

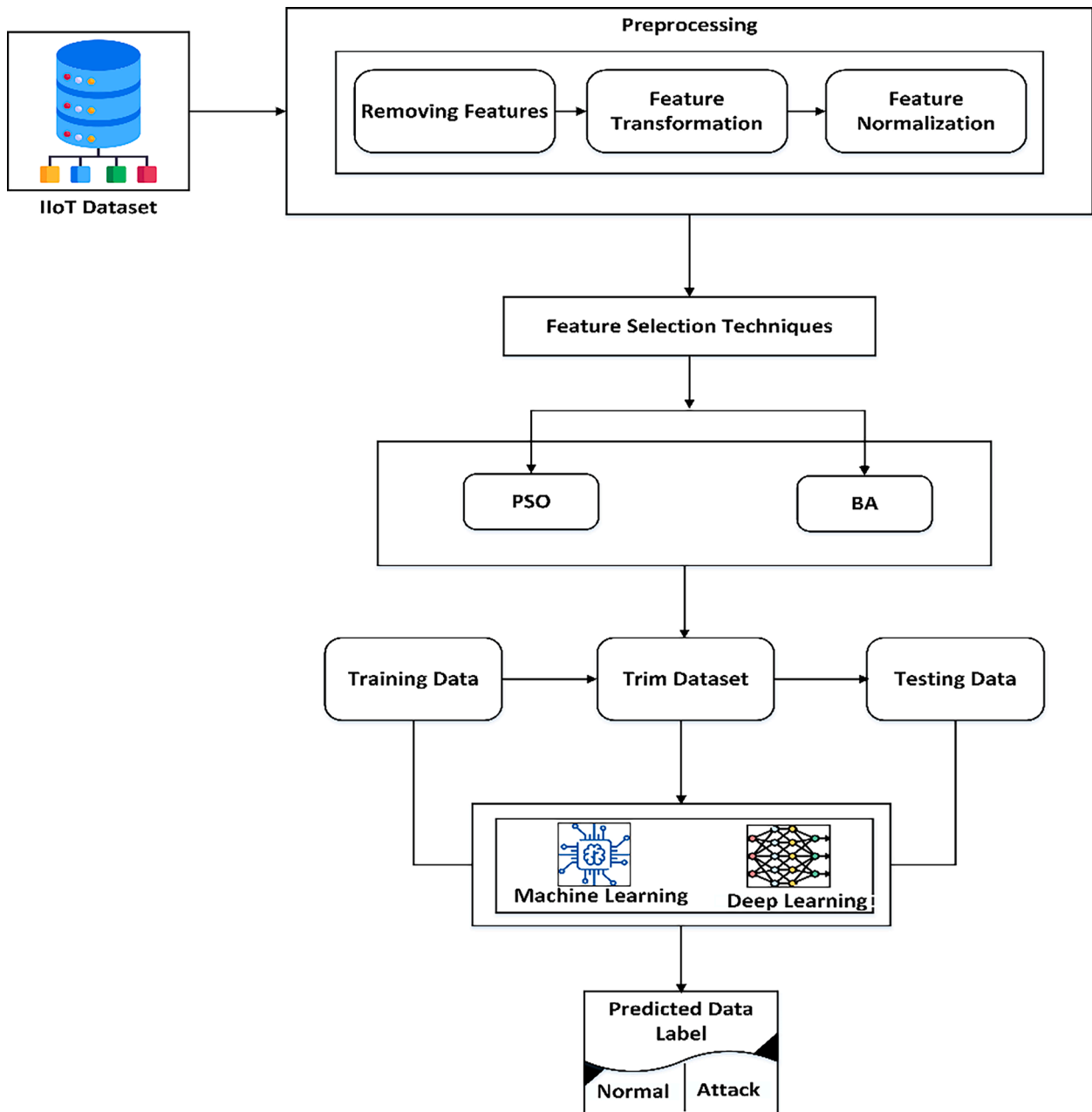


Fig. 3. Intrusion detection in IIoT.

programs referred to as smart contracts.

There are three distinct kinds of Blockchains that can be classified as Public, Private, and Consortium.

- i **Public Blockchain:** The permission less or public Blockchain is genuinely decentralized. The architecture allows any individual or device to partake in operations and engage as a validator within the consensus process. In public Blockchain systems, every block member can actively participate in block validation and maintain a copy of the Blockchain. Bitcoin and Ethereum, both open-source and smart contract Blockchain platforms, are the most renowned examples of public Blockchain systems. Public Blockchains, i.e., Ethereum and Bitcoin, are distributed public Blockchain platforms intended to serve many users [178]. The process of adding a new block to the Blockchain system involves significant computational effort. Each transaction incurs processing fees, which serve as rewards for peers. Miners utilize a proof-of-work (PoW) algorithm to achieve a secure

and tamper-resistant consensus across all network nodes. This algorithm verifies blocks and adds them to the Blockchain. The Blockchain is designed to be tamper-proof, making it extremely challenging and costly to alter the contents of any block [119]. Each block requires a particular quantity of gas, an Ethereum currency, to execute and pays miners as a reward.

- ii **Private Blockchain:** Permissioned Blockchain, commonly known as the Private Blockchain, is another name for the Private Blockchain, each network connecting to a node belongs to a single organization. In a permissioned Blockchain, the transactions on the Blockchain must be approved by a consensus mechanism before they can be written or retrieved; only authorized users/members can join and publish the new block. This Blockchain is free of transaction and token fees. Compared to the public Blockchain network, in contrast to public Blockchains, private Blockchain frameworks are not very resilient to temper-proof, and any business can restore its Blockchain network at any time. The Consortium Blockchains allow

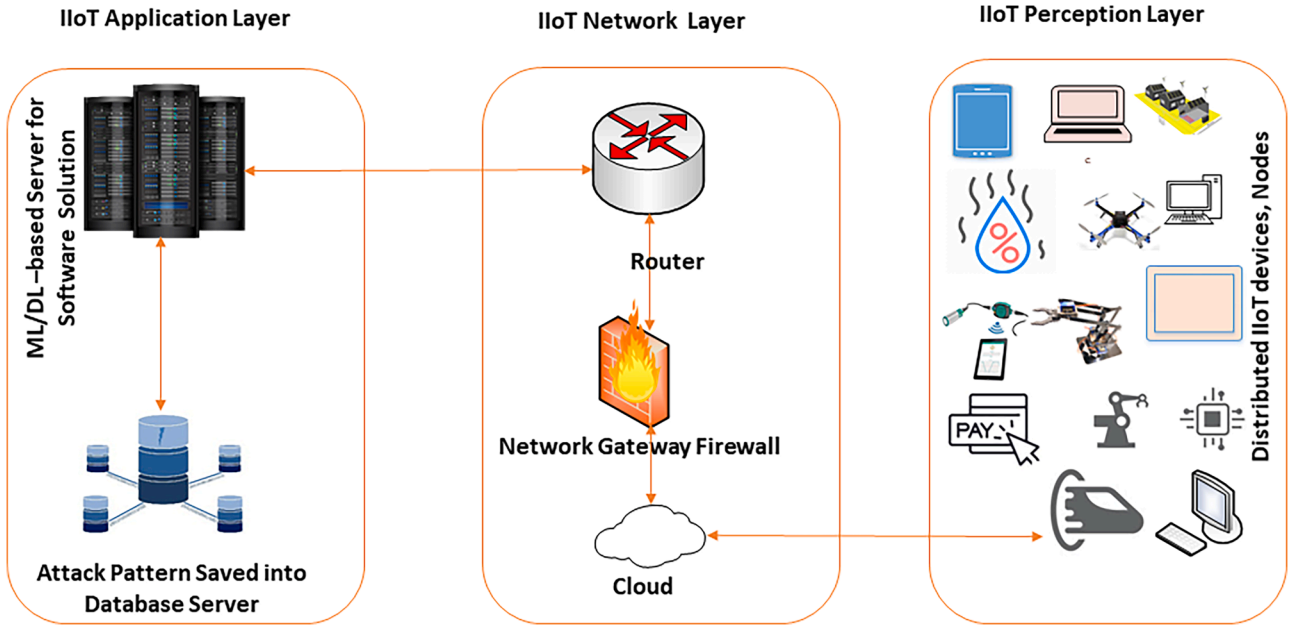


Fig. 4. An IDS network architecture with ML/DL and NGFW to detect, defend, and prevent known and new threats.

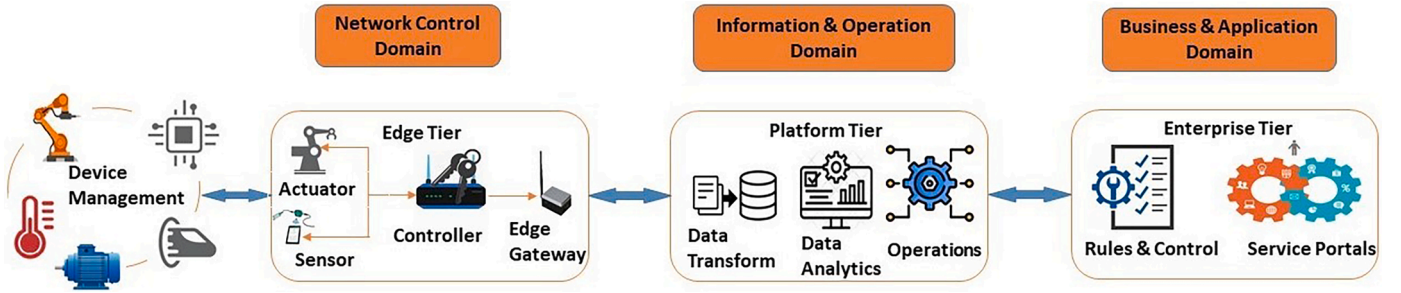


Fig. 5. Three tier IIoT system architecture.

organizations to change their business models and connect their operations because they can share control over the network. In addition to providing trust and scalability, a private Blockchain system removes the potential for a single point of failure and detects various attacks [153,12]. A Blockchain-based collaborative intrusion detection system (CIDs) is intended to secure Blockchain technology by enhancing accountability, consensus, integrity, resilience, scalability, and identity management to boost trust across numerous nodes.

- iii **Consortium Blockchain:** A distributed database that is auditable and constantly synced, known as a consortium Blockchain, is utilized to maintain track of data transfers between the consortium members. The consortium Blockchains are often adopted in scenarios where a shared ledger is required among a group of trusted entities, such as in supply chain management, healthcare systems, or financial consortia. The collaborative nature of consortium Blockchain allows for improved transparency, data sharing, and efficiency while maintaining control and privacy among the participating organizations. The Blockchain consortium is not used for processing, similar to private Blockchains. Furthermore, consortium Blockchains are hybrid Blockchain that integrates public and private Blockchain into a single operating system. For instance, Ripple [149] is a form of practical byzantine fault tolerance (PBFT) frequently utilized for Blockchains developed by consortiums. In contrast to other types of Blockchains consortiums, Blockchains have better scalability

because they are managed by a single entity or a collection of entities, and consensus can be reached rapidly.

3.3. Federated learning

Google introduced the concept of federated learning (FL) as a solution to address data privacy concerns by enabling collaborative learning across a diverse array of IoT devices [101]. The core concept of FL is the secure construction of machine learning models from distributed data sources and endpoints without allowing data leakage. In 2016, FL was introduced as a collaborative learning approach to overcome the limitations of traditional centralized ML methods. Unlike centralized approaches, FL ensures privacy by allowing end devices, known as clients or parties, to keep their data private. Instead of sharing raw data, clients only communicate partial model updates to a central entity known as an aggregator or coordinator. The aggregator aggregates these updates to create a global model, enabling collaborative learning without compromising data privacy. Fig. 7 shows a comparison between FL approaches, i.e., Centralized, Distributed, and FL. This innovative framework of FL has opened up new possibilities for secure and privacy-preserving machine learning in distributed environments [101]. In addition, to its application in cybersecurity for IoT, FL has been investigated in various domains, including smart cities, vehicular ad hoc networks, wireless communications [182], healthcare [176], recommender systems [64], edge networks [71], electric grids [140], and many others. One of the inherent advantages of the FL framework is its

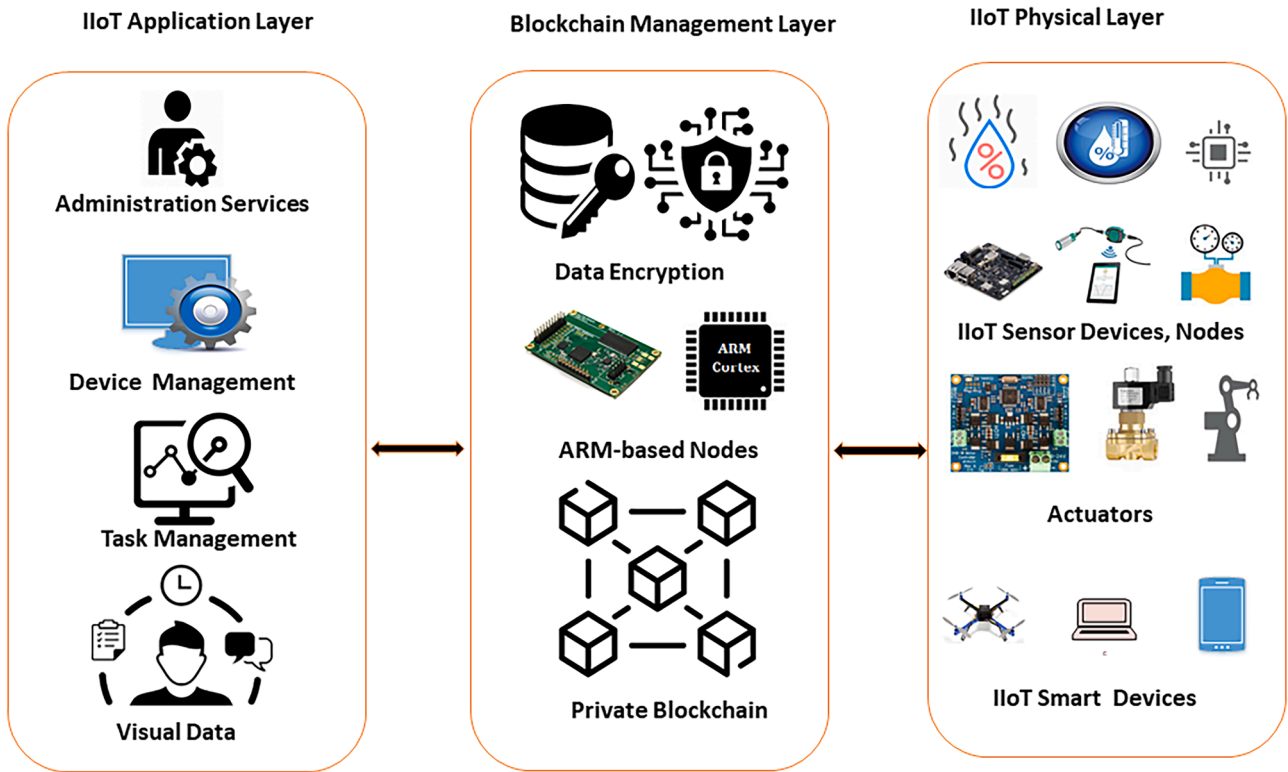


Fig. 6. The architecture for the IIoT using Blockchain technology.

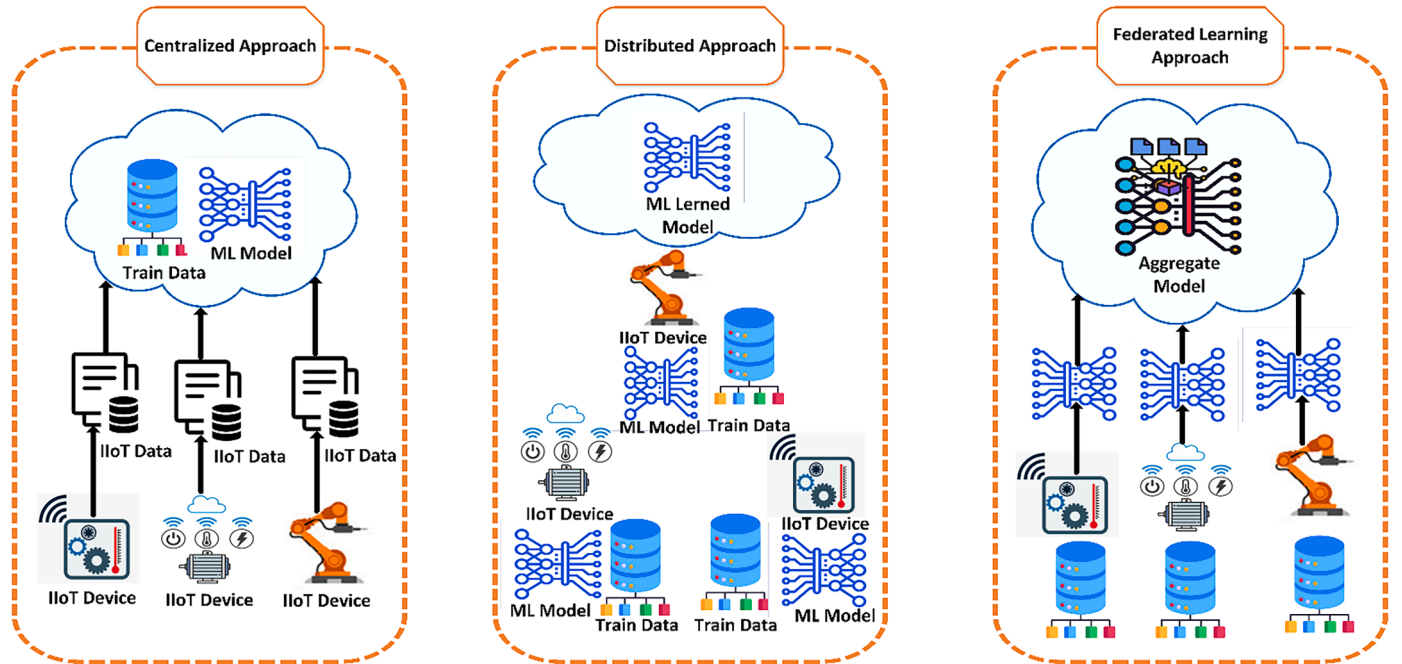


Fig. 7. Comparison between federated learning approaches.

support for security and privacy, which distinguishes it from the centralized learning framework.

In FL, data generated on an end device remains on itself, ensuring enhanced security and privacy protection. At present, researchers are integrating FL technology into IIoT to provide secure, accurate, resilient, and unbiased models [122]. FL allows different clusters to create a trained model using local data and share only the local model for aggregation. In contrast, the aggregated model is referred to as a global

model [81]. Many research studies are introducing Blockchain into their FL methodology since federated learning relies on a centralized server, which is vulnerable to attack. Additionally, to protect privacy and knowledge sharing, in model-driven development, instead of having clients send their data to a central server, the server sends models to clients so they may train on their local copies of the data. The authors in [79] comprehensively analyze FL-based IDSs. The study investigates the implementation of FL-based IDSs in different domains, elucidates

architectural distinctions, and provides an overview of the current state of the art in FL-based IDS from its inception in 2016 until 2021. The study establishes a reference architecture and a taxonomy that can be used as guidelines for comparing and designing FL-based IDSs. In their study Aydogan et al. [18] observed that centralized Intrusion IDS exhibited superior performance in detecting attacks in a timely manner. This advantage stems from their ability to collect and utilize a broader range of information compared to individual IDS agent nodes. Additionally, the centralized IDS approach was found to be well-suited for overcoming the resource limitations commonly encountered in IIoT settings. Fig. 8 present the general FL architecture.

Several studies have introduced FL-based approaches for IDS, as evidenced by existing works [135,158]. IDS plays a vital role in modern security but comes with distinct challenges. For example, detection systems must provide swift alerts to ensure timely responses [74]. They also need to handle diverse network settings and devices while effectively addressing unknown attacks [30]. Consequently, evaluating how various federation settings influence detection performance is crucial. Recent surveys have acknowledged the significance of FL-based IDSs in the research community by recognizing the interconnectedness of FL and IDS [26]. The authors in [51] proposed the study offers a comprehensive background and comparison of centralized learning, distributed on-site learning, and FL. Subsequently, it explores the application of FL in the context of cybersecurity for the IoT. While the primary focus of this study is on security, it also explores various approaches that tackle performance-related concerns (such as accuracy, latency, and resource constraints) associated with FL. These performance considerations are crucial as they can impact IoT systems' security and overall performance. FL provides an efficient anomaly-based IDS for restricted and distributed IIoT data, as well as decentralized IDS decision-making for heterogeneous IIoT architecture [53,136]. While all data is stored on the device, FL ensures the privacy of edge device data during the training phase. Furthermore, Smart edge devices can concurrently create mutual predictions with each other [143]. Blockchain is used to secure the legitimacy of locally obtained data and learned models during the FL

process. By providing a distributed ledger, Blockchain has paved the way for secure data sharing in IIoT [56]. However, FL faces many challenges in IIoT, such as, security threats on the servers, sensor nodes, and data breaches [11]. Considering this perspective, we believe it is imperative to present an up-to-date systematic review specifically focusing on IDS-based cybersecurity for IIoT. This review will shed light on the challenges and critical aspects that researchers must consider while developing an effective security system for IIoT. Reviewing the state-of-the-art to summarize existing knowledge is essential, and integrating Blockchain with FL-enabled IDS in IIoT networks can meet some challenges.

4. The threat to IIoT networks

IoT/IIoT devices have vulnerabilities that malicious actors can exploit for financial gain, data theft, or sabotage due to their interconnected nature and limited resources. Security attacks against IoT/IIoT exploit weaknesses in authentication, encryption protocols, and firmware vulnerabilities, leading to unauthorised access, device manipulation, data breaches, and network disruptions. Implementing robust security measures and risk assessments and collaborating on standards and best practices is vital to address these challenges. This approach will mitigate risks and support IIoT technology's continued growth and advancement.

There are two types of attacks that may happen to IIoT devices: external attacks, in which an attacker from outside the IIoT network attempts to compromise one of these devices, and insider attacks, in which a device within the network is used to compromise another. In threat modeling, STRIDE is typically used to classify and detect threat vectors. The following categories apply to the internal threat sources:

- T1—An intruder can stealthily access and record all communications inside a network and use this information for their ends [170].
- T2—An intruder can compromise a node. This vulnerability can potentially expose the entire Blockchain, enabling attackers to

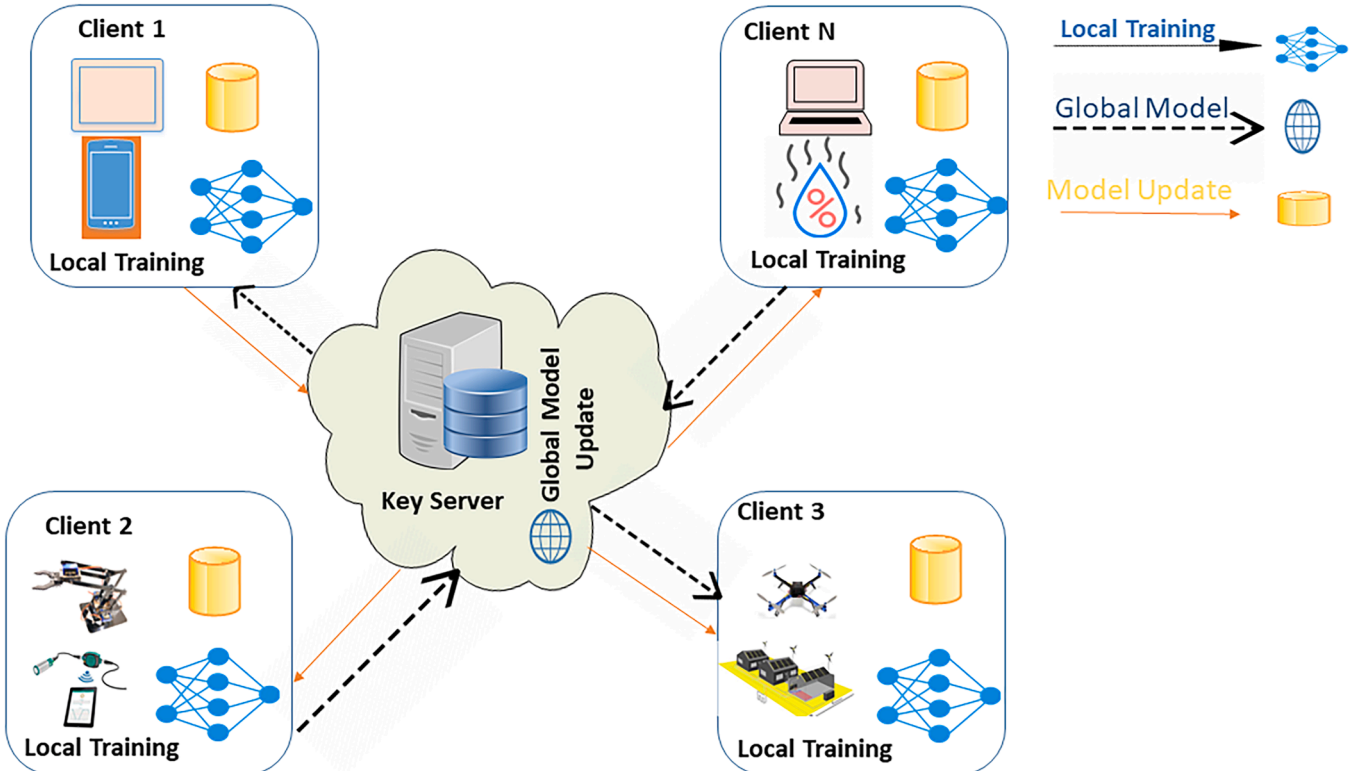


Fig. 8. General federated learning architecture.

continuously access the complete transaction and Blockchain history.

- T3—A malevolent attacker may attempt to exploit the vulnerability in previously deployed smart contracts.
- T4—An adversary with malicious intent might discover certain sensitive or secret information of the private credentials can be exposed in transactions or smart contracts, making them vulnerable to unauthorised access [175,172].

The majority of real-world attacks are highly targeted, trying to exploit specific vulnerabilities in IoT and IIoT devices and networks. Fig. 9 shows a high-level perspective of potential dangers affecting various parts of an IoT/IIoT system. This is critical since upgrading security features on a large number of sensors in the field is impractical, error-prone, and even damaging to users. Many attacks are aimed at exploiting zero-day vulnerabilities.

IIoT threat modeling generates measurable data that includes threats and associated risks. In order to counteract advanced persistent threats that are able to circumvent conventional security measures, one option is to implement a strong IDS throughout their lifecycle. We employed a risk model, STRIDE, which encompasses spoofing, tampering, repudiation, information disclosure, denial of service (DoS), and elevation of privilege to assess potential threats. The attacks on IIoT systems can be launched over an array of software sub-systems generally stacked into a layered architecture. The following subsection opens the discussion about such attacks using a layered-based taxonomy.

4.1. IoT attack taxonomy

This section specifically addresses the potential security threats that may occur within each of the four layers of the IIoT system. Fig. 10 provides a taxonomy of security attacks in the IIoT network according to these four layers.

4.1.1. Security considerations in the perception layer

The Perception Layer consists of diverse smart IoT devices, including

sensors and actuators, responsible for sensing the surrounding environment and collecting various types of environmental and industrial data [77]. It leverages diverse sensing technologies like GPS, RFID, WSN, and RSN. Addressing security challenges at this layer is crucial, as any compromise could severely affect the entire system. However, this layer is vulnerable to various attacks described below.

- **Node capturing attacks:** Node capturing attacks involve infiltrating or substituting a node within an IoT network or modifying hardware components. These nodes are susceptible to multiple cyber-attacks, presenting a significant risk to the overall IoT system. One such threat involves attackers capturing or substituting an IIoT system node with a malicious counterpart [57]. This type of malicious node poses a grave risk to the integrity and compromises the overall security of the entire IoT application.
- **Side-channel attacks:** The sensitive data within an IoT system is vulnerable to exploitation through various side-channel attacks. These attacks exploit the microarchitecture of processors, information obtained from power consumption analysis, and electromagnetic emanation to extract valuable information that can be utilized by an attacker [165]. Modern chips incorporate multiple security features to mitigate these risks and employ diverse cryptographic techniques to prevent and defend against potential side-channel attacks.
- **False data injection attacks:** In the IoT system, an IoT node can be tampered with to introduce fabricated data, inaccurate results or the malfunctioning of the IoT application [24]. Attackers may exploit this method as a means of conducting a DoS attack as well.
- **Eavesdropping and interference attacks:** In IoT scenarios involving the deployment of multiple nodes in open environments, there is an increased vulnerability to potential eavesdropping risks [86]. Consequently, malicious actors expose these applications to the potential interception of valuable data during the transmission or authentication phases.
- **Sleep deprivation attacks:** In these attack scenarios, perpetrators aim to deplete the power sources of edge devices; sleep deprivation

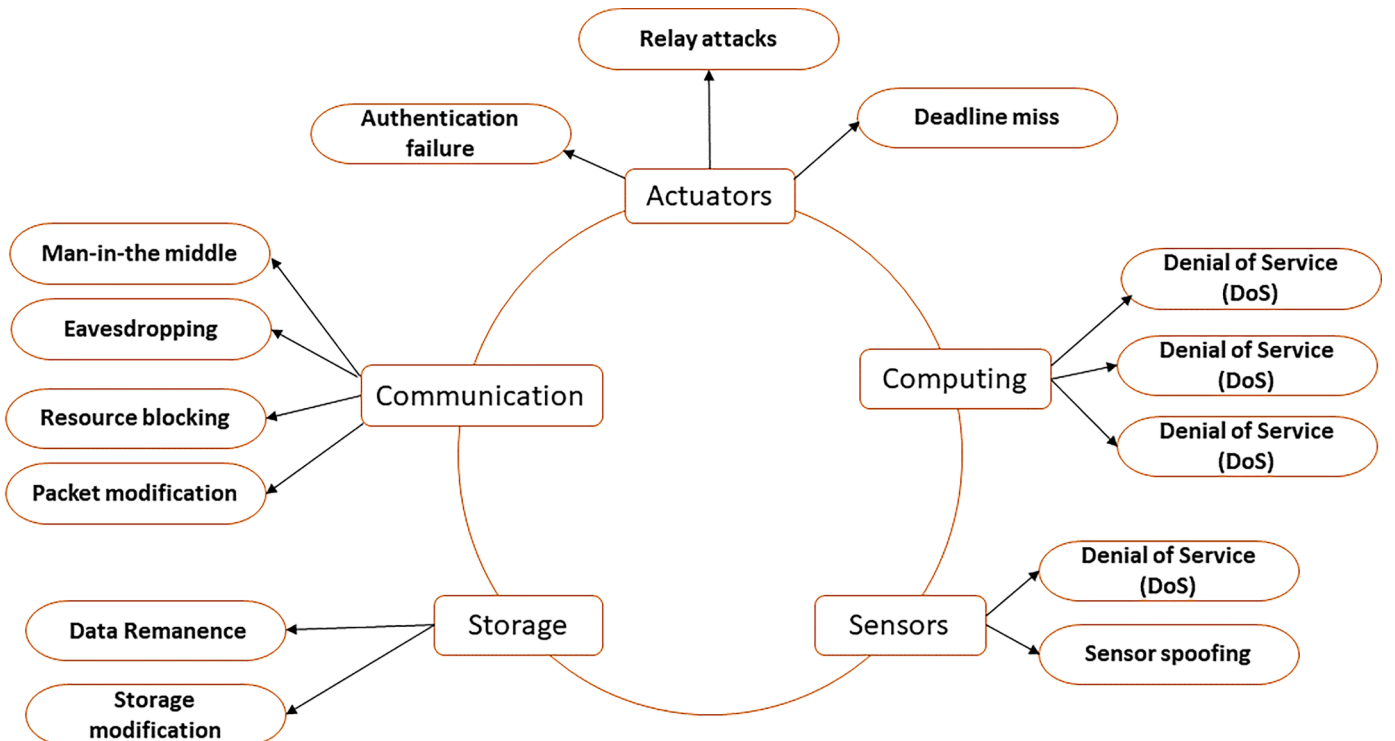


Fig. 9. Example attacks IIoT systems.

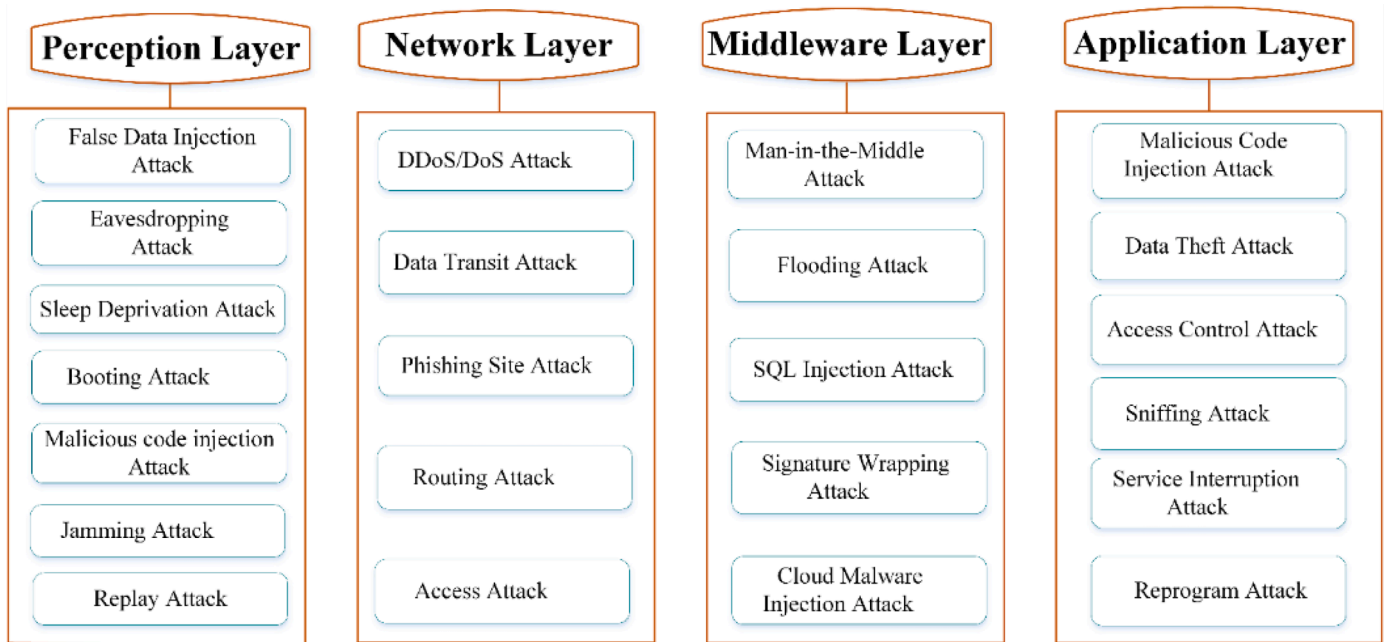


Fig. 10. Security attacks in the IIoT network.

attacks bear a resemblance to DoS attacks as they deplete the batteries of edge devices, conditions within the IoT nodes due to the unavailability of power sources [147]. This attack is executed by running malicious code in infinite loops on the edge devices, artificially escalating their power consumption.

- **Booting attacks:** IoT edge devices are susceptible to attacks due to the absence of security procedures during the boot stage [125]. Attackers can exploit these vulnerabilities by capturing the device during a reboot. In order to mitigate these attacks, it is crucial to establish a secure boot process for these devices.
- **Malicious code injection attacks:** These attacks commonly occur during over-the-air (OTA) software or firmware upgrades, which entails injecting malicious code into the memory of IoT nodes. It is a potential entry point for attackers to introduce malicious nodes into the IoT system [180]. During scheduled firmware updates, intruders may introduce Trojans into the system when devices are in operation mode, requiring a reboot. Addressing this security challenge involves two primary categories: authenticating and identifying edge devices to enhance network security, it is essential to prevent the installation of drivers or malware on peripheral devices under the pretense of updates or upgrades [76].
- **Jamming attacks:** These attacks involve disrupting or manipulating the tag reader air interface, which hinders or modifies communications. Jamming can be achieved through the use of a high-power, long-distance transmitter or through passive means such as shielding, taking advantage of the interface's sensitivity. RFID systems are susceptible to jamming by employing radio noise that matches the frequency of the system [76].
- **Replay attacks:** This attack type entails replicating an authentication code utilized by an authorized individual. It can be accomplished by cloning the authorized tag or intercepting and retransmitting signals captured from a smart device equipped through the communication antenna and signal card. The replay attacks demand access to particular data during the communication of the devices [76].

4.1.2. Security considerations in the network layer

The central role of this layer is to transmit information from the perception layer to computational units. Subsequently, we delve into the core security concerns associated with the network layer.

- **Dos/DDoS attacks:** Denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks involve the attacker inundating the network with excessive unwanted requests, rendering network servers inaccessible to legitimate users. DDoS attacks specifically employ multiple sources to flood the network with requests. While these attacks are not specific to IoT applications, the weak configuration of numerous IoT devices within the network can provide a vulnerable entry point for intruders to orchestrate such attacks against servers [61].
- **Data transit attacks:** The vulnerability to data breaches increases due to the diverse range of technologies used in data processing within IoT applications. Data transmitted between IoT devices becomes a prime target for intruders, while data stored in both cloud, edge and local servers faces security challenges. IIoT applications demand significant data processing and storage capabilities.
- **Phishing site attacks:** In such attacks, the strategy involves targeting multiple IoT devices, with the expectation that the attack will have an impact on at least some of them. The likelihood of such attacks is heightened when users visit unauthorised web pages. Once attackers access user accounts and passwords, the entire IoT system becomes susceptible to cyber-attacks [61].
- **Routing attacks:** This form of attack manipulates the routing paths within the IoT framework. During sinkhole attacks, nodes are deceived by false advertisements of the shortest routing paths, resulting in the redirection of traffic through the malicious path. On the other hand, warm hole attacks create unauthorised out-of-band connections between two nodes, facilitating the transfer of unauthorised packets. Both routing attacks undermine the fundamental security measures within an IIoT application.
- **Access attacks:** This attack is frequently referred to as an advanced persistent threat (APT). During this attack, an unauthorised intruder gains access to the IoT network and can persist undetected for an extended period. The primary goal of such attacks is to clandestinely steal valuable information from IIoT applications. Due to the data-intensive nature of IIoT applications, their data-processing activities are particularly susceptible to these persistent attacks.

4.1.3. Security considerations in the middleware layer

The layer plays a pivotal role in enabling communication between

the application and network layers by acting as a mediator and offering advanced computing and storage features [133]. The middleware layer is susceptible to various attacks, especially concerning the security of databases and cloud and edge environments.

Man-in-the-middle attacks: An attacker in an IIoT network may intercept and alter the communication between two parties, which is known as Man-in-the-Middle (MitM) attacks. This scenario may result in unauthorised entry, data tampering, or the compromise of confidential information. In the context of the MQTT protocol, which utilizes a publish-subscribe communication model, subscribers and clients are decoupled from each other [3]. If an attacker successfully executes a MitM attack, it can lead to unauthorised access, data manipulation, or the extraction of sensitive information within the IIoT system.

- **Flooding attacks:** This attack functions similarly to a Denial-of-Service (DoS) attack and significantly impacts the quality of service. The attacker continuously inundates a service with a large number of unwanted requests, leading to an overload on cloud servers and making the service inaccessible to legitimate users [130].
- **SQL injection attacks:** The intruder has the ability to inject malicious SQL statements into the main program, enabling attackers to gain unauthorised access to private user data or manipulate valuable data stored in the database [96]. According to the open web security project (OWASP), SQL injection attacks have been categorized as one of the most critical web security threats.
- **Signature wrapping attacks:** Within the middleware layer, web services utilize XML signatures. However, an intruder can compromise the security mechanism and manipulate intercepted messages by exploiting vulnerabilities in the object access protocol [138].
- **Cloud Malware injection attacks:** These attacks involve an intruder's injection of malicious code or virtual machines into cloud servers. The attacker can gain access to sensitive user information by impersonating a legitimate service using a virtual machine [50].

4.1.4. Security considerations in the application layer

The application layer plays a pivotal role in formatting and presenting data while delivering a wide range of IIoT applications to end-users. These applications encompass various domains such as smart health, homes, cities, industries, and transportation. In this layer, security emerges as a central challenge, leading to recurrent concerns [16]. The following section describes the primary security issues encountered in this layer.

- **Malicious code injection attacks:** Adversaries tend to exploit the most rudimentary and readily accessible methods to initiate attacks on a system. In the absence of robust code validation mechanisms, the system becomes susceptible to malicious scripts and potential vulnerabilities. Cross-site scripting (XSS) is a common technique employed by intruders, where they inject malicious scripts into trusted websites. An adeptly executed Cross-Site Scripting (XSS) attack can lead to severe consequences, such as the compromise of user accounts through hijacking or even causing significant disruptions to the entire IIoT system [157].
- **Data theft attacks:** In IIoT applications, the transmitted data is more vulnerable to cyber-attacks than data at rest, given its vulnerability during transit. Exploiting data theft attacks in such applications can have severe implications for registered users [157]. To counteract these attacks effectively, robust security measures can be implemented, encompassing advanced encryption techniques, network authentication mechanisms, and data isolation protocols.
- **Access control attacks:** Access control is a crucial authentication mechanism that permits legitimate users to access accounts and process data. In IIoT applications, compromising access control poses a significant threat. Upon successful compromise of access, the entire infrastructure becomes exposed to a myriad of potential threats, leaving it vulnerable to a wide range of security risks [181].

- **Sniffing attacks:** Adversaries possess the capability to employ sophisticated sniffer applications for the purpose of monitoring network traffic in IIoT systems. Therefore, it is imperative to establish and deploy resilient security measures within the IIoT infrastructure to effectively thwart any unauthorised access and safeguard confidential user data [4].
- **Service interruption attacks:** These attacks, known as disruptive interruption attacks, strategically disrupt the availability of services to authorized users. Through intentional overloading of servers, they incapacitate their responsiveness, ultimately leading to service unavailability [80].
- **Reprogram attacks:** If the programming process of IIoT objects lacks adequate protection, an attacker may attempt to reprogram them. This can potentially result in the complete hijacking of the IIoT network, giving adversaries significant control over its operations and critical components [3].

4.2. Threat to federated learning as IDS/IPS engine

The requirement to safeguard many devices from unauthorised access while maintaining system stability and providing prompt responses in real time has complicated the threat problem in the industrial setting. The ability to identify threats in IIoT devices is crucial in this context, as it allows for the elimination of potentially harmful factors such as injection by rogue IIoT devices. The difficulty of threat modeling has increased due to the proliferation of new threats and the various variants developed from existing ones. Effectively modeling traffic and providing a means of detecting both previously-known and as-yet-undiscovered threats depends on having access to robust security mechanisms. The evident risk lies in the openness of a machine learning model to manipulation, which can occur when a mere handful of edge devices function erroneously. The Attackers may take advantage of vulnerabilities in federated learning systems utilized in IIoT networks for the implementation of the IDS/IPS systems.

Following are the vulnerabilities an adversary can exploit in FL-based systems in IIoT/IIoT networks.

4.2.1. Poisoning attack

The main goal of adding false data into the aggregation process, poisoning attacks primarily aim to decrease a model's predictive power. Nevertheless, according to the origins of the trained updates, the authors in [160] classified poisoning attacks as either involving the poisoning of models or data. In data poisoning, the training data on compromised edge devices is altered, whereas in model poisoning, updates to the poisoned model are generated using a set of predetermined rules. Using trimmed optimization with multiple keys, the authors in [93] suggested a secure federated learning process immune to numerous common poisoning threats.

4.2.2. Information leakage

Hitaj et al. [60] outlined a novel approach involving the implementation of deep learning models, specifically a generative adversarial network (GAN), to craft a set of specialized training patterns that are resilient against adversarial attacks. However, the process of training a deep neural network across an extensive dataset might take a substantial amount of time and resources. Based on an investigation of the privacy leakage the authors in [173,38] suggested a safe and resilient federated learning protocol for IIoT networks called East Fly. In their proposed framework, a local node receives parameters from the central server at regular intervals during the training process. Utilizing its own training dataset, the local node computes necessary modifications and directly communicates these updates back to the server. Moreover, the server modifies the parameters and then make changes to the overall settings based on what it receives. Collaborative learning is vulnerable to severe inference cyberattacks, as discovered by Melis et al. [102] because of the leaking of unwanted characteristics.

4.2.3. Denial-of-service (DoS) attack

DoS attacks aim to disrupt or overwhelm the availability of IIoT systems by flooding them with excessive traffic, requests, or malicious data. This can lead to system downtime, causing operational disruptions and financial losses.

4.2.4. Malware and ransomware attack

Malware and ransomware attacks involve installing malicious software or code within IIoT systems. Malicious software, or malware, has the capacity to cause operational disruptions, exfiltrate sensitive data, and grant unauthorized access to attackers. In particular, ransomware employs encryption to lock vital data, demanding a ransom in exchange for its decryption, thus inflicting considerable disruptions and imposing substantial financial consequences.

4.2.5. Byzantine attack

To subtly alter the max-model predictor's classification outcome, an attacker can employ a byzantine attack to send out locally-modified harmful models to other participants. The adversary can also corrupt their local copy of the model update process. Using Blockchain technology, the authors in [169] built a secure federated learning infrastructure that can withstand Byzantine attacks. It is a notion to use a digital twin with adaptive federated learning proposed by [158]. Its foundation is based on interaction records and educational efficacy; both rely on deploying damaging updates to mitigate the threat posed by malicious data.

4.2.6. Jamming attack

A sophisticated jamming attack targeting FL-based security and privacy solutions endeavors to deliberately disrupt the normal functioning and communication of the system. with or clash with the receiver to obstruct communication inside the victim network. Using federated learning, Mowla et al. [115] advocated a safety framework for detecting cognitive jamming attacks. However, using the client group prioritizing method, on the Dempster-Shafer theory as its foundation, detection may be carried out on the device while accounting for the environment's uneven sensory data characteristics.

4.2.7. Shilling attack

In a shilling attack, an attacker tries to manipulate recommendation systems by creating a significant number of phony profiles. The ultimate goal of a shilling attack is to manipulate a recommendation system by artificially increasing or decreasing the popularity of a specific product or service by creating a large number of fake user profiles and giving those profiles unreasonable ratings. The authors in [67] provided a novel system for detecting fraudulent attackers by developing four characteristics from gradient matrices; the proposed solution involves training a semi-supervised Bayes classifier.

4.2.8. Zero-day exploits

Zero-day exploits capitalize on undisclosed and previously unknown vulnerabilities present within IIoT systems, for which no patch or remediation is currently available. Attackers exploit these vulnerabilities before they are discovered or patched, potentially causing significant damage and compromising system security.

4.2.9. Privacy leakage attack

Differential attacks allow an attacker to determine whether or not an IIoT has participated in a mission based on local model modifications. Some information about the training data can be leaked even after a distributed learning approach has been implemented, primarily due to the presence of up-to-date local model parameters on IoT devices. The authors in [169] introduced an innovative methodology that leverages a combination of Blockchain technology, local differential privacy, and reinforcement learning as a formidable defense mechanism against the data leakage. Data leakage attacks in federated learning expose

participant data through a breach in the aggregate server caused by malicious behavior. Additionally, ML models may be attacked in numerous ways, including white-box, black-box, and gray-box assaults. Flooding in a white box assumes the vulnerable target can be easily accessed [186]. Adversaries employing black-box attacks can solely interact with the network through output queries, without any substantial knowledge of the network's internal structure or parameters. These three approaches may often be grouped under the umbrella of adversarial attacking methods. The study provides proof of concept for the efficacy of bolstering IIoT network cyber security by integrating Blockchain and FL.

5. State-of-the-Art IDS approaches for IIoT networks

5.1. Blockchain-based approaches for IIoT-ID

Blockchain is a decentralized, immutable ledger. It removes a vulnerable central server in an untrusted computer environment [46]. Blockchain provides an efficient method from a IDS perspective, which is useful in designing security solutions for IIoT. Each node in a Blockchain network operates as a peer-to-peer (P2P) node, and there is no requirement for additional protocols for P2P communication between nodes [19]. The implementation of an IDS ranks among the foremost and cutting-edge embraced cybersecurity solutions, representing a prevalent and extensively employed method for detecting and preemptively countering malicious activities that may arise within a network infrastructure.

A primary function of an IDS is to monitor both inbound and outbound network traffic, audit data, and security logs; it will send an alarm to the network management center if it detects any abnormal data packets [98]. Notwithstanding, Blockchain stands as an adaptable technology that fosters trust and reliability in decentralized and distributed systems. It has many potential applications, including those in IoT, healthcare, and supply chain management [97]. Since Blockchain can protect data integrity and ensure process integrity, it can potentially be used in IIoT for intrusion detection [151]. The IDS is designed to detect anomalous activity. In addition, IDS systems may also be used to ensure a safe IIoT system [168].

Furthermore, known security threats that take advantage of zero-day vulnerabilities are one of the threats that most frequently affect IIoT networks. The attackers utilize a number of methods, such as denial-of-service (DoS), decentralized DoS (DDoS), and Spoofing, to infect and modify the behavior of targeted devices. Examples include the 2010 Stuxnet worm assault on Iran's nuclear program and the 2013 intrusion by Iranian hackers into the dam's industrial control system (ICS) in New York. However, another example of a cyberattack took place in Ukraine, where over 80,000 blackouts in 2015 due to a passive black-energy strike [41,17]. Signature analysis, anomaly detection, and specification-based analysis are the three primary types of detection used in an IDS system. Intruder detection systems that use either signatures or anomalies effectively thwart cyber assaults [190]. The increasing sophistication of cyberattacks renders the use of a single IDS insufficient for protecting a network. Contrarily, CIDs are employed to provide a robust platform where isolated IDS nodes may communicate and share data on harmful activity discovered within their subnetworks while minimizing false positives [10,98]. IIoT-based systems are vulnerable to attack because centralized systems do not have adequate security and insufficient trust amongst CIDs, making it difficult for IDS to successfully thwart cyber-attacks in IIoT systems. Hence, securing IIoT environments against potential security risks involves creating a reliable and holistic security monitoring system and an effective IDS to cope with novel cyber-attacks in IIoT systems.

The distributed and verifiable nature of Blockchain technology makes it possible for IIoT systems to link previously untrusted devices. Fig. 11 shows the framework for Blockchain-based IIoT systems and its components. The construction of a Blockchain comprises three primary

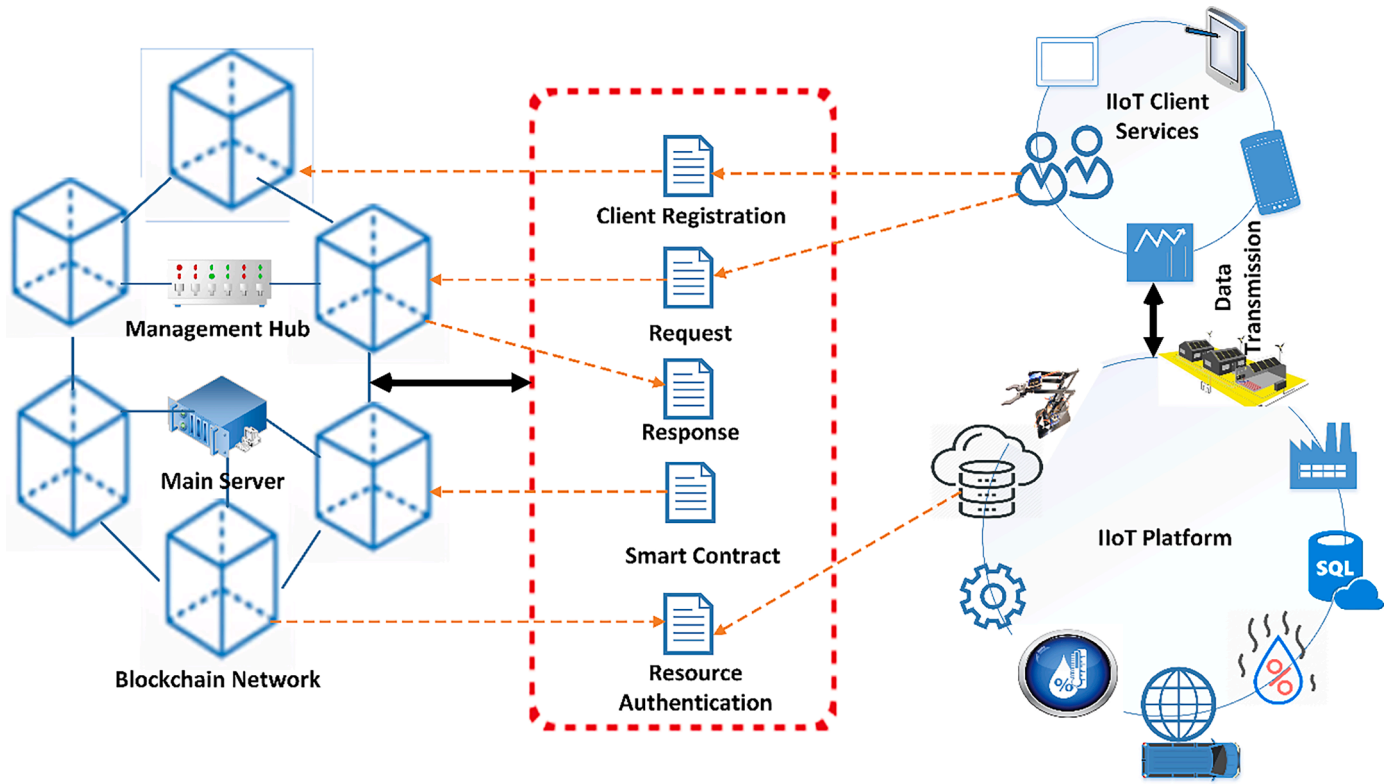


Fig. 11. Blockchain-enabled IIoT architecture.

components; hash value, Hash Function/Timestamp, Block body, and Merkel root.

- A hash value: The hash value is responsible for storing data from the prior block.
- Hash function/Timestamp: The Blockchain data is stored in the timestamp and hash function, which records the exact moment new blocks are created together.
- Block body: During the block's creation, the block body retains every single examined transaction. Construct a Blockchain, and these individual blocks are connected by hash values.
- A Merkel root: This results in fewer transactions being processed within each block, making it easier to verify the existence and integrity of transaction data quickly.

The architecture of a Blockchain-enabled IIoT system, consists of the six components listed below.

- 1) Blockchain-enabled Network: It uses a private, decentralized network to store all the data in the systems.
- 2) IIoT Networks: The IIoT system is able to supply these various resources to its users.
- 3) The Key management Servers: It generates the appropriate cryptographic keys for node authentication and data encoding.
- 4) Management Hub: It focuses primarily on managing and maintaining the entire system.
- 5) Smart Contract: It supports IIoT-Blockchain system interfaces.
- 6) Clients: Users who make access requests to the IIoT resources fall under this category.

IIoT modernizes smart industries by incorporating the newest technologies. Integrating Blockchain technology into the IIoT platform raises the level of security and attracts significant interest from academia and business.

5.2. Federated learning-based approaches for IIoT-ID

While maintaining data privacy, the authors in [136] presented an FL approach that relies on client and server interaction using a deep neural network (DNN) to develop an IDS model. Similarly, the authors in [59] developed an FL detection approach that emphasized communicating aspects such as specific threat information to each alert. With IDS, machine learning (ML) and deep learning (DL) have gained tremendous momentum in achieving high accuracy and classification. Additionally, IDS employing ML approaches learns from normal and abnormal data traffic by executing the model's training process. Federated Learning was presented in 2016 to identify the malicious flow of data. Through a series of iterative training sessions, a group of clients or parties on the network's periphery train an AI algorithm, and then that data is sent back into the network's core system to improve the global model [101]. FL is a decentralized learning method that protects privacy and trains models locally before sending parameters to a central server. The FL has recently come into consideration as a conceivable approach for learning a shared model collectively from patterns, dataset, or environment distributed over a network [5].

Prior to distributing its up-to-date global model to clients for each iteration, the FL server strategically selects a subset of clients to engage in the learning process [110]. Secondly, FL empowers devices to engage in collaborative learning without necessitating centralized data exchange. Moreover, various machines and servers can train ML/DL using decentralized data in multiple rounds [159]. Next, the clients/users update the FL server with the new model or learned parameters. Upon receiving an updated global model or learned parameters, each client or user utilizes their data. The known model can be stored in the Blockchain using the FL method to improve how well data is stored and keep data private. To employ safe, accurate, resilient, and impartial models, researchers have utilized FL technology in IIoT [23]. However, FL still faces challenges, i.e., security risks, while implementing and using FL in IIoT, including data leakage and server and client-side attacks [62]. On the other hand, integration of Blockchain-enabled with FL in the IIoT

system faces some adoption challenges. Furthermore, to give additional privacy protection, decentralization, and security in IDS, Blockchain-enabled FL can incorporate ML/DL with IIoT networks during the classification and training phase.

FL's system architecture comprises three primary IDS deployment architectures that can be used according to the system's needs. These architectures are Centralized architecture, Decentralized architecture, and Distributed architecture. IDS are typically deployed centrally for small networks with limited scalability. The main reason behind the success of FL is because of its collaborative machine learning model's decentralized training process; FL is able to significantly reduce the burden on the centralized server in terms of both computing and storage costs, in addition to protecting the privacy of sensitive information shared across several user interfaces and application frameworks. This architecture requires data transmission to a central server to train the ML model for IDS. More than one IDS actively identifies attacks in a decentralized and distributed architecture. To enhance the capabilities of the IDS, the authors in [142] integrate inter-agent communication and hierarchical decision-making mechanisms. However, the effectiveness of these features is hindered by the restriction of data being stored locally. Fig. 12 shows the different IDS types that can be deployed in FL architectures.

FL offers a server-client architecture that includes both server-side computations, such as model aggregation, and client-side interaction, i.e., model training. The authors in [136], employed Multi-Layer Perceptron (MLP) and auto-encoder models for IDS systems. They used statistical analysis to compare results from centralized and distributed FL-based systems. Furthermore, IDS deployment can be adapted to cater to both large-scale and small-scale networks, encompassing network-based IDS for broader monitoring, host-based IDS for individual system protection, or protocol-based IDS targeting specific communication protocols. These variations cater to the diverse needs of different applications and end-users. In a conventional FL system, both the central authority and the trainees/participants have a high degree of autonomy. Using the FL system, researchers in [62] have created models that are safe, precise, robust, and impartial for IIoT use cases. When developing large-scale distributed systems, FL is often used to build centralized models without compromising data privacy to enhance training performance [22,46,156]. Consequently, IDS in IIoT networks would be more trustworthy if combined with Blockchain and FL-enabled IDS to establish a robust line of defense against security and privacy attacks.

5.3. Hybrid IDS approaches

Recently, with its billions of IIoT devices, highly dispersed data exchange, and decentralized data interaction, edge-based IoT presents new

issues in building an applicable IDS. To ensure the security of the IIoT, IDS design is essential for IIoT networks. Building an effective IDS for edge-based IIoT requires both the detection technique and the system architecture as fundamental building blocks. Generally, an IDS helps prevent harmful traffic since it is used to identify network activity, notably to differentiate between normal and malicious traffic. When creating detection methods and system architecture for IIoT devices, resource utilization efficiency should be addressed due to each small edge device's inadequate storage and computing ability. The hybrid IDS approach combines the benefits of Blockchain and FL-enabled in IIoT networks, using various detection techniques among lower- and upper-layer network devices, respectively. Blockchain has become the industry standard for securing data transfers between nodes in a distributed network [164]. Blockchain is utilized to enhance security in distributed IDS networks due to its immutability and the consensus procedures followed by nodes. The majority of currently available security applications share multimedia files and sensitive data using Blockchain technology [20]. Several types of security applications employ Blockchain technology to increase performance. Blockchain technology has lately been utilized in a wide range of fields, including IoT, IIoT, IDS, and financial services, as well as various other domains [85,144,132]. Currently, there is a growing concern about the deployment of Blockchain in IIoT Networks [84], for example, to protect data transfers by integrating Blockchain technology with FL, ML, and Deep Reinforcement Learning (DRL), the sharing mechanisms of IDS are optimized, thereby enhancing data collection efficiency and overall efficacy [88].

The implementation of FL for IDS paves the way for DL/ML to deliver individualized protection mechanisms across many devices. As a security tool, federated learning has been proven effective. Conventionally, IDS have been considered crucial to keeping systems safe by spotting possible security attacks/threats based on traffic monitoring and analysis. IDS is responsible for identifying and detecting malicious network flows to prevent the progress of network attacks and log the network flow information for further research. Addressing core privacy, data security, and digital rights management concerns, FL empowers multiple participants to collaboratively build robust machine learning models without the need for direct data exchange. The current state of data-sharing systems that rely on a central curator increases the likelihood of data leakage, especially when multiple parties are involved. The demand for IDS is increasing since IPS-based network attack prevention is no longer feasible owing to the emergence of fresh forms of assaults in IIoT systems on a daily basis. The use of FL in various aspects of anomaly detection is investigated together with different IDS models, ML approaches, and related difficulties. In addition, there are numerous approaches, including neural networks and clustering algorithms, now being taken into consideration in recent years regarding its use in IDS [89,28,40,179]. In the environment of the IIoT, current initiatives have

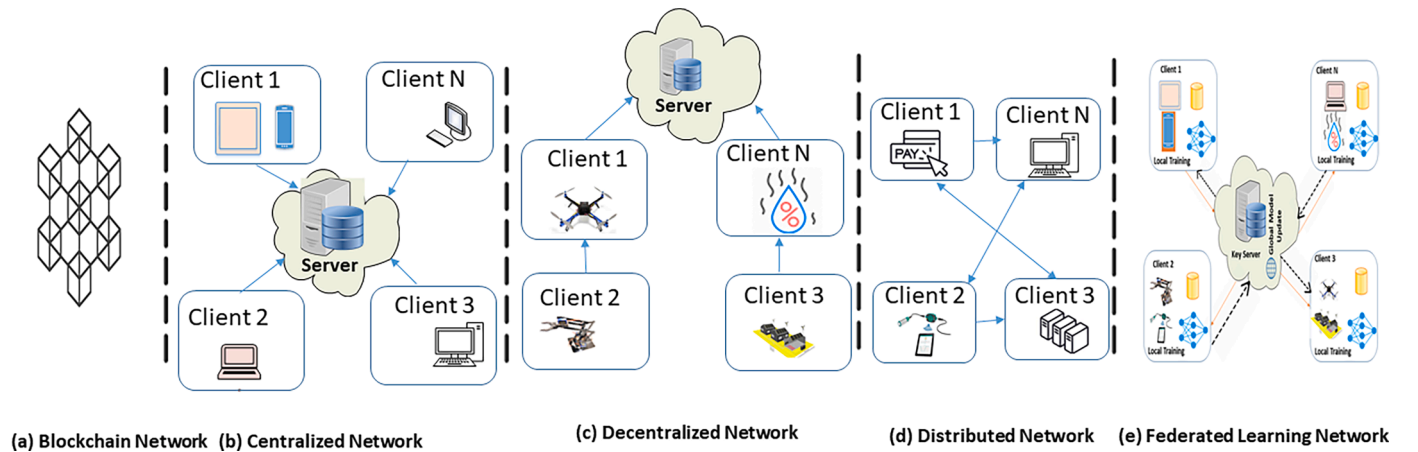


Fig. 12. Different IDS architectures deployed in FL.

been presented by taking into consideration various technologies and devices related to the security in [32]. The authors in [48,47] assess DL approaches for detecting network security threats. Despite the recent major advancements in IDS for IIoT, there is still much potential for improvement in developing effective IDS [6,154]. To this end, the creation of cutting-edge distributed IDS systems is urgently required, such as Blockchain and Federated Learning-enabled to be developed for IIoT networks.

5.4. Intrusion detection datasets

Assessing an IDS's efficacy necessitates utilizing a dependable and up-to-date dataset encompassing contemporary instances of legitimate and anomalous activities. The dataset serves as a vital resource for accurately evaluating the performance and effectiveness of the IDS in detecting and distinguishing between normal and suspicious behaviors. The datasets enable researchers and cybersecurity experts to train, validate, and test IDS algorithms to distinguish between legitimate network traffic and malicious intrusions accurately. Table 2. presents the summary of well-known datasets used in the literature for experimentation of intrusion detection systems. Over time, numerous datasets have been generated, and while many of them are kept private to address privacy concerns, a subset has been made accessible to the public. Below, we provide concise explanations of the commonly employed datasets in IoT/IIoT settings.

5.4.1. KDD-Cup99

The dataset used for the experiment was derived from the DARPA-98 dataset, specifically designed for evaluating the performance of IDSs that can effectively differentiate between inbound attacks and normal network connections. The KDD-Cup99 [70,27] is widely acknowledged and accepted as a benchmark dataset in the field. The dataset represents a modified version of the DARPA-98 dataset, initially supported and funded by the Defense Advanced Research Projects Agency (DARPA), as part of an IDS evaluation plan conducted at MIT's Lincoln Laboratory [100]. The main purpose of this dataset was to assess the performance of IDSs in distinguishing between inbound regular network activity and malicious activities. The dataset contains 41 attributes similar to a Net Flow dataset. However, KDD-Cup99 has numerous weaknesses, discouraging its use in the current context [163].

5.4.2. NSL-KDD

The NSL-KDD dataset was introduced as an innovative remedy to address the limitations identified in the KDD99 dataset. The development of the NSL-KDD dataset was undertaken with the explicit objective of rectifying shortcomings and bolstering the efficacy of intrusion detection evaluations within the domain of cybersecurity [148]. The dataset represents a balanced resampling approach to the KDD-99 dataset, focusing on examples likely to be overlooked by classifiers trained on the original KDD-99 dataset. Nevertheless, the dataset addresses some of the limitations, and it is important to note the authors themselves acknowledge specific weaknesses, such as the dataset's lack of illustration of low-footprint attacks [100,131]. In the IDS realm, diverse technologies and classification algorithms are employed to effectively develop a robust IDS framework utilizing this particular data set under consideration.

5.4.3. UNSW_NB15

The UNSW – NB15 dataset [114], originating from the University of New South Wales Australia's Defense Force Academy, is a pivotal resource within the domain of cybersecurity, has gained significant popularity in the research community as a prominent benchmark for IDS in the context of IoT. With a total of 80,873 samples, the UNSW_NB15 dataset is a valuable resource for researching network intrusion detection. Within this dataset, there are 19,488 instances of normal IoT traffic with 49 features. Simultaneously, the remaining 61,385 entries

encompass ten distinct categories of attacks, namely Analysis, Backdoor, DoS, Exploit, Fizzers, Generic, Reconnaissance, Shellcode, and Worm. The development of the UNSW-NB15 dataset was undertaken by the Australian centre for Cyber Security (ACCS), representing a significant advancement in the realm of cybersecurity aimed to address the limitations of previous intrusion datasets and facilitate the detection of emerging cyberattacks, including those with low footprints. Utilizing the IXIA Perfect Storm tool, they generated a substantial 100 GB dataset, incorporating a mix of modern network traffic and synthesized data. This synthesis involved the inclusion of both normal and abnormal traffic within a controlled laboratory environment [114].

5.4.4. Bot-IoT

The Bot-IoT dataset [112] was meticulously curated within the Cyber Range Lab at the center of UNSW Canberra Cyber, with a specific emphasis on constructing a lifelike network environment that accurately represents the intricate IoT ecosystem. This dataset offers an extensive collection of records featuring diverse network profiles. Researchers also introduce a testbed configuration to address limitations found in existing datasets, such as the absence of comprehensive network information, accurate labeling, and coverage of the latest and complex attack variations. They employ various machine learning and statistical techniques for forensic purposes. The Bot-IoT dataset comprises an extensive collection of over 72 million network traffic records, obtained from a simulated IoT environment. It encompasses a diverse array of attacks, including OS and Service Scan, DoS, DDoS, Data exfiltration, and Key-logging. Furthermore, the author offers a downscaled variant of the dataset, containing approximately 3.6 million records, meticulously designed for the sole purpose of conducting comprehensive evaluations and assessments. However, the labeling files are systematically arranged, categorizing attacks based on their respective attack categories and subcategories. The dataset includes original pcap files, generated argus files, and CSV files, each contributing to the comprehensive analysis and assessment of the data [75].

5.4.5. TON_IOT

The TON_IOT dataset [15], an advanced initiative of the IoT Lab at UNSW Canberra Cyber, School of Engineering and Information Technology (SEIT), Canberra at the Australian defence Force Academy (ADFA), offers a comprehensive and sophisticated compilation of data from various sources, spanning the domains of both IoT and IIoT [15]. This dataset contains telemetry labeled IoT/IIoT data originating from multiple virtual machines, implementing diverse operating systems facilitates the establishment of cross-layer connectivity, interconnecting the IIoT, Cloud, and Edge/Fog systems in a comprehensive and advanced manner. The dataset involves a wide variety of nine major IoT traffic types, providing a comprehensive representation of network activity, operating system logs, and other IIoT service traces. The TON_IOT dataset further includes several attack scenarios, such as Man in the Middle, injection, DoS, DDoS, backdoor, password cracking attacks (PWA), ransomware, scanning, Cross-Site Scripting (XSS), and cross-site scripting attacks, enriching its utility for research and analysis in the domain of IoT and IIoT security [113].

5.4.6. X-IIoTID

The X-IIoTID dataset was created at the University of New South Wales in Canberra with the aim of simulating the tactics, techniques, and procedures employed by emerging attackers within IIoT environments [8]. The dataset incorporates a wide range of IoT components, including sensors, actuators, controllers, edge devices, mobile devices, and cloud-based systems. The dataset covers a wide range of communication styles, including Machine-to-Human, Machine-to-Machine, and Machine-to-Machine interactions, characterized by high network activity and numerous events. Additionally, various communication protocols like CoAP, WebSocket, and MQTT are included. The dataset provides a comprehensive representation of the heterogeneous network

Table 2

An evaluation of recent IoT cybersecurity datasets available for intrusion detection.

Dataset	Year	IoT traffic	Class	Records percentage		Features	Description	Traffic feature
KDD-CUP99	1999	Normal	Normal	972,780	19.85 %	41	The KDD99 dataset was derived from the DARPA 98 dataset to serve as an assessment tool for intrusion detection systems aimed at distinguishing between incoming attacks and regular network connections.	IoT IIoT
		Attack	Dos	3,883,370	79.27 %			× ×
			U2R	50	0.001 %			
			R2L	1126	0.02 %			
			Probe	41,102	0.83 %			
		Total		4,898,428	100 %			
NSL-KDD	2009	Normal	Normal	972,781	19.85 %	41	NSL-KDD dataset, which is a widely used benchmark dataset in the field of intrusion detection and network security. It is an improved version of the original KDD Cup 1999 dataset, commonly used to evaluate IDSs.	× ×
		Attack	Dos	3,883,366	79.27 %			
			Probe	41,102	0.83 %			
			R2L	1126	0.02 %			
			U2R	52	0.001 %			
		Total		4,898,427	100 %			
UNSW_NB15	2015	Normal	Normal	19,488	24.10 %	49	The dataset is derived from a synthetic environment created to generate attack activities, serving to evaluate ensemble intrusion detection models.	✓ ×
		Attack	Generic	39,496	48.84 %			
			Exploits	16,187	20.02 %			
			DoS	1791	2.21 %			
			Fuzzers	1731	2.14 %			
			Reconnaissance	1703	2.11 %			
			Analysis	264	0.33 %			
			Worms	114	0.14 %			
			Backdoor	114	0.12 %			
		Total		80,873	100 %			
Bot-IoT	2019	Normal	Normal	9543	0.001	29	Bot-IoT incorporates legitimate and simulated IoT network traffic and various types of attacks used by botnets.	✓ ×
		Attack	Dos					
			TCP	12,315,997	1.672 %			
			UDP	20,659,491	2.81 %			
			HTTP	29,706	0.004 %			
			DDoS					
			TCP	19,547,603	2.66 %			
			UDP	18,965,106	2.58 %			
			HTTP	19,771	0.002 %			
			Reconnaissance					
			OS Fingerprinting	358,275	0.04 %			
			Service Scanning	1463,364	0.19 %			
			Information Theft					
TON_IOT	2020	Normal	Normal	3086,973	85.60 %	31	The dataset is employed to evaluate FL techniques in the context of intrusion detection.	✓ ×
		Attack	Backdoor	246,136	6.83 %			
			DDoS	53,992	1.50 %			
			Injection	50,319	1.40 %			
			Password	142,674	3.96 %			
			Ransomware	16,030	0.44 %			
			Scanning	3973	0.11 %			
			Xss	6037	0.17 %			
		Total		3606,134	100 %			

(continued on next page)

Table 2 (continued)

Dataset	Year	IoT traffic	Class	Records percentage		Features	Description	Traffic feature
X-IIOTID	2021	Normal	Normal	4,211,417	513.07 %	68	The dataset comprises device-agnostic data utilized in the context of ML/DL learning based on IDS for both IoT and IIoT systems.	✓ ✓
			Attack					
		Attack	Generic Scanning	50,277	6.13 %			
			Scanning	52,852	6.44 %			
			Vulnerability					
			Discovering	23,148	2.82 %			
			Resources					
			Fuzzing	1313	0.16 %			
			Brute Force	47,241	5.76 %			
			Dictionary	2572	0.31 %			
			Insider Malicious	17,447	2.13 %			
			Reverse Shell	1016	0.12 %			
			Mitm	117	0.01 %			
			Modbus Register	5953	0.73 %			
			MQTT	23,524	2.87 %			
			TCP Relay	2119	0.26 %			
			Command Control	2863	0.35 %			
			Exfiltration	22,134	2.70 %			
			Ransomware	458	0.06 %			
			RDoS	141,261	17.21 %			
			Fake Notification	28	0.00 %			
			False Data Injection	5094	0.62 %			
		Total		820,834	100 %			
WUSTL-IIOT	2021	Normal	Normal	797,261	92.71 %	41	This dataset was developed by incorporating authentic data from diverse IIoT and industrial devices, including both legitimate and malicious inputs; this application aims to simulate a real-world IIoT environment accurately.	✓ ✓
		Attack	DoS	56,379	6.56 %			
			Reconnaissance	5932	0.69 %			
			Command Injection	185	0.31 %			
			Backdoor	152	0.25 %			
		Total		859,927	100 %			
Edge-IIoTset	2022	Normal	Normal	1380,858	71.65 %	61	A real-time testbed-based cyber security dataset for IoT and IIoT applications is employed to assess intrusion detection systems.	✓ ✓
		Attack	DDoS_UDP	121,567	6.31 %			
			DDoS_ICMP	67,939	3.53 %			
			SQL_injection	50,82	2.64 %			
			DDoS_TCP	50,062	2.60 %			
			Vulnerability_scanner	50,026	2.60 %			
			Password	49,933	2.59 %			
			DDoS_HTTP	49,203	2.55 %			
			Uploading	36,915	1.92 %			
			Backdoor	24,026	1.25 %			
			Port_Scanning	19,983	1.04 %			
			XSS	15,066	0.78 %			
			Ransomware	9689	0.50 %			
			Fingerprinting	853	0.04 %			
			MITM	358	0.02 %			
		Total		1927,304	100 %			

traffic and system activities encountered in IIoT environments, thus holding significant value for cutting-edge security research. With 68 attributes, it covers a wide spectrum of attack types, including reconnaissance, weaponization, Ransom Denial of Service (RDoS), exfiltration, crypto-ransomware, and tampering. The dataset comprises a total of 820,834 data records, of which 421,417 are labeled as normal instances and 399,417 as attacks [15]. By providing network traffic data from diverse IIoT devices and protocols, the X-IIoTID dataset offers researchers a prevailing tool to enhance the security of IIoT systems and devise effective defense mechanisms against emerging threats [37].

5.4.7. WUSTL-IIoT

In 2021, Maede Zolanvari from the University of Washington's McKelvey School of Engineering introduced the WUSTL-IIOT dataset, strategically designed to bolster cybersecurity in IIoT environments [191]. This dataset was meticulously crafted by modeling and emulating real-world industrial systems, ensuring their relevance and practicality. The WUSTL-IIOT dataset comprises authentic network traffic data captured from diverse IIoT devices. It encompasses 1107,448 normal samples and 87,016 samples with occurrences of DoS, Reconnaissance, Command Injection, and Command Injection attacks, spanning a total of 41 characteristics. The dataset's generation involved the implementation of a supervisory control and data acquisition (SCADA) architecture

within the IIoT testbed. With a specific focus on intrusion detection, the WUSTL-IIoT dataset is a valuable resource for researchers seeking to develop and assess IDS algorithms and systems tailored explicitly for IIoT applications. This dataset enables the evaluation of AI/ML-based IDS, primarily employing binary classification to determine whether a given traffic sample indicates an attack or not, thus contributing to the advancement of IIoT security measures [36,121].

5.4.8. Edge-IIoTset

The Edge-IIoTset is a state-of-the-art and comprehensive dataset specifically designed for IoT/IIoT environments [44]. The dataset was meticulously compiled through a rigorous seven-layer test that incorporated over ten distinct types of IoT devices. These devices comprise temperature and humidity sensors, ultrasonic sensors, water level detection sensors, pH sensor meters, soil moisture sensors, heart rate sensors, and flame sensors. These devices were incorporated into an IIoT-based Modbus flow, covering a range of IoT/IIoT protocols. Tailored explicitly for edge computing in IoT and IIoT settings, the Edge-IIoTset dataset aims to analyze and categorize threats effectively. It encompasses 14 different attack types, including information gathering, injection attacks, man-in-the-middle attacks, and malware attacks. The dataset comprises 1380,858 instances of normal IoT traffic and 546,446 recorded attacks, offering valuable insights for ML and DL-based IDS. These systems can utilize the Edge-IIoTset dataset in two distinct modes: centralized and FL. To guarantee the dataset's relevance and applicability, a specialized IoT/IIoT testbed was employed. This testbed incorporated a wide variety of devices, sensors, protocols, and cloud/edge configurations, all of which were utilized to create the dataset. This includes capturing data pertaining to alerts, system resources, logs, and network traffic, with a novel addition of 61 new features demonstrating high correlations among the existing 1176 features [37].

6. Discussion

To deliver a reliable and trustworthy solution for the IIoT networks, we integrated Intrusion Detection Systems using Blockchain and Federated Learning-enabled approaches in our survey. Table 3. summarizes several state-of-the-art IDS techniques integrated with FL and Blockchain in IIoT networks. The table presents a variety of intrusion detection approaches, security threats, performance metrics, time complexity, and deployment specifics, demonstrating how integrating FL and Blockchain technologies enhances the security of IIoT networks. Even though IDS is an essential component in the IIoT network security stack to detect prevalent attacks but still poses a limitation on the detection of novel and newer network attacks; the research in IDS to address the limitation focused on dynamic approaches such as ML-based intrusion detection systems (ML-IDS) in IIoT. ML-IDS analyzes IIoT network traffic for anomalies to discover new attack types by learning typical network behavior from previously recorded data packets. However, we argue that ML-IDS deployed today still weakens emerging new attack types. The scope and scale of the system monitored by the ML-IDS do not have the oracle knowledge of ever-changing anomalous network data flows.

Furthermore, ML-IDS works by building previously recorded IoT devices' network traffic which cannot ensure prevention against attacks in case of a cold start or new deployment. In addition, it is known that the performance of ML-IDS varies for a particular network environment due to the dynamics of the data captured by the IDS [92]. This variation can be interpreted and visualized from the reported results of ML-IDS techniques on different network traffic datasets. Thus, developing a holistic, centralized ML-IDS is not trivial, and sharing (centralizing) network flow information across other organizational networks poses severe security and privacy concerns.

Recently, considerable research on ML for intrusion detection in information and communication system networks has been reported.

Many conventional centralized ML techniques, for instance, support vector machine (SVM) [124,146], and ML (DL) [56,68,120], reinforcement learning (RL) state reasonable achievements for detecting malicious user activities in IoT and IIoT systems.

FL is an innovative paradigm for distributed collaborative learning and computing explicitly designed for machine learning-based applications [45]. Recent research has shown that FL-based IDS might serve as a viable security paradigm to address the requirements and problems in IIoT security. Li et al [81] proposed an FL technique that uses GRU and CNN for determining the presence of incursions in the CPS. In their work participants gain knowledge from the experiences of their peers through the transmission; the participants can fine-tune the performance of the IDSs by sending local gradients to a distant server for global gradient precomputation and sharing. In [87], unsupervised DL models are used for attack detection to address a significant amount of unlabeled data available in IoT and IIoT networks.

Similarly, to demonstrate an unsupervised FL architecture for detecting Android malware is proposed in [159]; this architecture aims for the collaborative training of a decentralized GAN network to mitigate and be resilient against anomalies in the data-gathering at a central server. The authors in [129] proposed mechanisms for protecting user data, preventing intrusion, and identifying attacks correlated to FL are discussed. The study in [150] designed a collaborative intrusion detection system (CIDS) for VANETs in which many SDN controllers work together to train a unified IDS model for the whole network without sharing details about the traffic patterns inside their individual segments. Moreover, an effective anomaly-based IDS against restricted and dispersed IIoT data is one of the many benefits of FL's decentralized IDS decision-making against limited and diverse IIoT infrastructure. [42, 136]. In contrast, the traditional IDS schemes may not apply to the ever-changing landscape of the IIoT due to their inherent adaptability.

Blockchain solves most of IIoT's security and privacy issues by guaranteeing consensus, immutability, and provenance; Blockchain adds credibility to IDS, wherein IDS nodes must be authorized before taking part [98]. IDS has become necessary to ensure the safety of recent IIoT networks and technologies. There are three widely acknowledged approaches for detection, signature-based, anomaly-based, and specification-based, utilized by intrusion detection systems. Since breaches into the IIoT nearly invariably include humans, expensive industrial goods like chemicals, and machinery, early identification of incursions is essential.

Considerable efforts have been devoted to developing new ML/FL and Blockchain-based IDS/IPS systems. Our research indicates that a federation of network intrusion detection systems that use Blockchain as a fabric of the federation and the principles of federated learning for network intrusion detection have to be considered for a robust, open, and adaptive IDS ecosystem. In the following subsection, we present our recommendation and guidelines recorded as a holistic research gap from the literature for implementing a Blockchain-assisted federation of intrusion detection systems (BF-NIDS).

6.1. Recommendation for blockchain and federated intrusion detection systems

Based on our literature survey and the critical analysis of the reported articles, we outlined the following pinnacle guidelines for Blockchain and Federated Network Intrusion Detection System (BF-NIDS):

6.1.1. Define the objective and scope of the federation

Clearly define the objectives and scope of your NIDS, including the types of network intrusions you aim to detect and the level of security you want to achieve. This will help determine the system's appropriate design and features [141]. Similarly, clearly define the scope of the federation, including the participating organizations, their roles, and the level of information sharing and collaboration allowed. This step helps

Table 3

Comparison of different state-of-the-art intrusion detection methods.

Reference	Year	FL- Based IDS	Blockchain IDS	IDS approach	Security threat	ML/DL method	Deployment	Dataset	Performance metrics%	Time complexity
[1]	2022	✓	✓	Cyberattack detection using a temporal convolutional generative network (TCGAN) trained on partially annotated data.	Cyber-attack detection	Semi-Supervised, GAN	Decentralized	ToN.IoT LITNET-2020	Acc = 95.89	—
[95]	2022	✓	×	FL, DL, Over-sampling approach with partially homomorphic encryption, Paillier Encryption	Web spam detection, Metadata spoofing	XGBoost LightGBM STM CNN	Decentralized	XIIoTID	Acc = 99.79	—
[9]	2022	✓	×	Deep-learning-based intrusion detection system for industrial IoT	Reconnaissance, Backdoors, DoS, and command injection, pose serious threats to system security and integrity	MLP, DeepIIoT	NIDS	WUSTL-IIOT	Acc = 99 Pr = 99 Rc = 99 F1 = 99	<u>Train Time</u> 25.4487 s
[44]	2022	✓	×	A realistic testbed was developed to evaluate the efficacy of machine learning-based IDSs in the IoT/IIoT environment.	DoS/DDoS, Information gathering, Injection, Man in the Middle, and Malware	SVM, DT, RF, KNN, and LSTM	Hybrid	Edge-IIoT	Non-IID = 99.98 IID = 100	—
[82]	2021	✓	×	IMA-GRU protocol	DDoS Attack	GRU	Decentralized	UNSW NB-15	Acc = 98	<u>MR-Time</u> 1715.91 s/ 1000bots <u>MD-Time</u> 483.74s
[110]	2021	✓	×	Implemented the LSTM and gated recurrent units (GRUs)	×	LSTM GRU	Decentralized	Real-time	Acc = 95	<u>Train Time</u> 45.31 s
[8]	2021	×	×	Standard ML and DL Algorithms	Network attack, Brute force attack, Dictionary attack, Reverse shell, MitM attack	NB, DT,SVM, KNN, LR, GRU DNN	Decentralized	X-IIoTID	Acc = 99.54 Pr = 97.20 Rc = 97.27 F1 = 99.49	<u>R-Time</u> 1049 ms
[63]	2021	✓	×	Lock edge is a cloud-based security system that protects data at the network's edge (Low-Complexity Cyberattack Detection in IoT Edge Computing)	DDoS (HTTP, TCP, UDP), DoS (HTTP, TCP, UDP), server scanning, OS fingerprinting, keylogging, and data espionage	NN, CNN, RNN, KNN, SVM, KNN, RF, DT,NN	Hybrid	BoT-IoT	Acc = 99	<u>CPU Usage</u> 400 to 2400/s <u>Attack Rate</u> 9.60 sample/s
[193]	2021	×	×	Proposed a universal,	Backdoor, Command	AI, ML	NIDS	WUSTL-IIOT, NSL-	Acc = 99.98 Acc = 99.24	<u>T-Time</u> 212.65s

(continued on next page)

Table 3 (continued)

Reference	Year	FL- Based IDS	Blockchain IDS	IDS approach	Security threat	ML/DL method	Deployment	Dataset	Performance metrics%	Time complexity
				accurate, and well-performing XAI model called TRUST XAI, AI-based trustworthy systems for IIoT	injection			KDD, UNSW NB-15	Acc = 97.77	
[69]	2021	×	×	Proposed an IDS for IIoT using the Genetic Algorithm (GA)	Access control attacks, data corruption breaches, spoofing attacks, DoS, DDoS, OS, Jamming attacks	RF, LR, NB, DT, ET, XGB.	NIDS	UNSW-NB15	Acc = 87.61 VAC = 95.87 Rc = 98.34 Pr = 82.51 F1 = 89.73 Auc = 98	<u>Test-Time</u> 18.3 ms
[90]	2021	×	×	Proposed a particle swarm optimization-based gradient descent (PSO-LightGBM) for IDS.	Backdoor, Shellcode, and Worms	LightGBM, OCSVM	NIDS	UNSW-NB15	Acc = 86.68	<u>Train Time</u> 12.502 s <u>Test Time</u> 0.048s
[159]	2020	✓	×	The Federated Generative Adversarial Network (F-GAN) and the Generative Adversarial Network (GAN)	Poisoning Attacks	Unsupervised GAN	Decentralized	Drebin Genome, Contagio (Android malware)	Acc = 97.96	—
[128]	2020	×	×	Create a collaborative IDS using deep learning and SDN to detect abnormal network behaviors across the entire VANET infrastructure.	GPS spoofing attacks, Sybil attacks, timing attacks, wormhole attacks, DOS attacks, hidden vehicle attacks	DL, Bi-GAN, distributed SDN.	Hybrid	KDD99 NSL-KDD	Acc = 96 – 98 Pr = 91– 95 Rc = 91 – 96 F1 = 91 – 95 Auc = 97 – 98	—
[55]	2020	×	×	The study proposed an adaptive trust-boundary protection model that uses deep-learning-based semi-supervised approaches.	MitM attacks, Function code injection, Firmware modification	DL-based Semi-supervised Learning	NIDS	Real-time	Acc = 96.63 Auc = 96	—
[15]	2020	×	×	ML and DL Methods in both binary and multi-class classification problems for intrusion detection	Scanning, DoS, DDoS, Ransomware, Backdoor, data injection, Cross-site Scripting (XSS), and MitM	SVM, KNN, RF, DT, LDA,NB, LR	Hybrid	TON_IoT	Acc = 88 Pr = 90 Rc = 88 F1 = 88	<u>Train Time</u> 6.308 s <u>Test Time</u> 0.022
[120]	2019	✓	×	FDL is used as autonomous self-learning distributed system for detecting compromised IoT devices.	Benign and Network Traffic Attacks	Unsupervised (FDL) GRU	Decentralized	Private IoT Smart Home	Acc = 98.2	<u>T-Expr</u> = 10 500/s <u>M-Time</u> 7.0 ms

(continued on next page)

Table 3 (continued)

Reference	Year	FL- Based IDS	Blockchain IDS	IDS approach	Security threat	ML/DL method	Deployment	Dataset	Performance metrics%	Time complexity
[192]	2019	×	×	Carrying out the backdoor, command injection, and SQL injection attacks on a system secured by an ML-based anomaly detection system.	Buffer overflow, Code injection, Improper input validation, SQL injection attack,	ML-based IDS	NIDS	Real-time	Acc = 99.99	—
[35]	2019	×	✓	Random Subspace Learning (RSL), K-nearest neighbor (KNN), and ensemble learning (EL) (KNN)	Misrouting attacks, Forged ICS commands, and SDN-related attacks	LSVM, Bayes Network BN, NB-K, KNN, AdaBoostM1, Bagging, DT, RF	Centralized	Real-time	Acc = 96.73	<u>Training Time</u> 3738 Instances KNN = 0.01–0.01 s Binary + M- class RSL- KNN=0.05–0.5 s Binary + M- class
[116]	2018	×	×	Feed-forward neural network (FNN), semi-supervised neural network (SNN), and auto-encoder (AE) (DFFNN)	Network attacks, zero-day attacks	Semi-Supervised AE+FNN	Centralized	NSL-KDD UNSW- NB15	Acc =98.6 Dr = 99 – 93 Fpr = 1.8 – 8.2 ROC = 98.4 – 92.5	<u>NSL-KDD =</u> <u>Train</u> 0.13 s/100ep <u>NSL-KDD =</u> <u>Test</u> 0.06 <u>UNSW-15 =</u> <u>Train</u> 0.25–0.14 s/ 100ep <u>UNSW-15 =</u> <u>Test</u> 0.01 s
[185]	2018	×	×	The proposed scheme utilizes denoising autoencoders (DAE) combined with a weighted loss function and a compact multilayer perceptron (MLP) for intrusion identification.		MLP, DAE	NIDS	UNSW- NB	Acc = 98.80 F1 = 95 Pr = 95.98 Rc = 94.43	—

MR-Time: Mitigation Response Time, MD-Time: Mitigation Delay Time, R-Time: Response Time, T-Expr: Total Experiment, M-Time: Attack Mitigation Time, Binary + M-class: Binary and Multi-Class, Acc: Accuracy, PR: Precision, Rc: Recall, F1: F1-Score, Auc: Area Under Curve (ROC), s: Seconds, ms: Milliseconds.

establish trust and sets the foundation for effective communication and cooperation.

6.1.2. Select a blockchain platform

Gain a solid understanding of blockchain technology, including its underlying principles, consensus mechanisms, and smart contracts. This knowledge will help you design an effective and secure blockchain-based NIDS. Furthermore, more importantly, choose a suitable blockchain platform that aligns with your requirements. Consider factors such as scalability, consensus mechanism, security, development community, and available smart contract capabilities. Popular choices include Ethereum, Hyper ledger Fabric, and Corda [88].

6.1.3. Design of distributed NIDS architecture

Develop a robust NIDS architecture that integrates blockchain technology in a federated learning setting. Consider the Sensor Nodes, Blockchain Networks, Smart Contracts, Federated Learning Model Aggregator, and User Interface [177].

6.1.4. Establish trust relationships

Establish trust relationships between participating organizations and components of the architecture by defining policies, agreements, and legal frameworks. It is important to ensure that the organizations involved have a common understanding of security objectives, data handling practices, and privacy concerns.

6.1.5. Standardize data formats and protocols

To enable effective information sharing, standardize the data formats, protocols, and interfaces used for communication between the participating IDS systems. This allows for seamless integration and interoperability among different IDS platforms [127].

6.1.6. Define intrusion detection rules

Develop intrusion detection rules or signatures based on known attack patterns, anomalies, or behavior-based techniques. These rules will be used to detect and classify network intrusions.

6.1.7. Data collection and storage

Determine how network traffic data will be collected and stored on the blockchain. Consider the balance between data privacy and system performance. You may choose to store aggregated metadata or anonymized data while maintaining the original traffic details off-chain.

6.1.8. Feature selection and engineering

Identify relevant features that effectively differentiate between normal and malicious network traffic flows. Consider elements like destination and source IP addresses, port numbers, protocol types, payload content, packet size, and packet header contents. If necessary, engineer new features, such as calculating statistical properties or aggregating information [141].

6.1.9. Model selection

In order to develop an effective network intrusion detection system, it is crucial to select the most suitable machine learning algorithm. There are several viable options available, including decision trees, random forests, neural networks, deep neural networks, and deep reinforcement learning, among others. The selection of a model can also be influenced by the characteristics of your data capture available to the federated clients, such as the number of features, sample size, and class imbalance when selecting the model [2].

6.1.10. Integration and deployment

Integrate the trained model into your network infrastructure, ensuring it can process incoming network traffic in real-time. Consider using specialized hardware accelerators or distributed systems for efficient processing in high-speed networks. Implement proper logging and alerting mechanisms to notify security personnel in case of a detected intrusion [112].

6.1.11. Define data sharing policies

Develop clear policies regarding what types of data will be shared, at what level of detail, and with whom. Consider the sensitivity of the data and privacy requirements, ensuring that personally identifiable information (PII) or sensitive data is appropriately protected [91].

6.1.12. Ensure data anonymization and privacy

Implement mechanisms to anonymize or pseudonymize sensitive data before sharing it across the federation. This protects the privacy of individuals and organizations while still allowing for effective analysis and correlation of security events.

6.1.13. Establish a centralized management entity

Designate a central authority or management entity responsible for coordinating the federation activities, managing the shared data, and ensuring compliance with established policies and regulations. This entity could also handle incident response coordination and facilitate participant communication.

6.1.14. Share threat intelligence

Encourage the sharing of threat intelligence among participating organizations. This includes sharing information about emerging threats, new attack vectors, and known indicators of compromise

(IOCs). Timely and effective sharing of threat intelligence improves the overall security posture of the federation [99].

6.1.15. Collaborative analysis and response

Enable collaborative analysis and response capabilities among the participating IDS systems. This can involve sharing aggregated analysis results, collaborating on incident response efforts, and sharing remediation strategies and best practices [142].

6.1.16. Continuous monitoring and evaluation

Implement continuous monitoring and evaluation mechanisms to assess the effectiveness of the individual component IDS and the whole federation of network intrusion detection system. Regularly review the federated system's performance, reliability, and security, and refine the policies and procedures based on lessons learned and evolving threats [142].

6.1.17. Compliance with legal and regulatory requirements

Ensure the federation of IDS meets all relevant legal and regulatory requirements, it is essential to comply with data protection and privacy laws, with GDPR (General Data Protection Regulation) being a crucial consideration. Consider jurisdictional differences and cross-border data-sharing restrictions while designing and operating the federation [13]. Implementing a BF-NIDS requires careful planning, collaboration, and adherence to established guidelines and best practices. By leveraging multiple organizations' collective knowledge and resources, BF-NIDS can significantly enhance the security and resilience of network infrastructures.

6.1.18. Challenges and open issues

Developing a dependable detection system is one of the greatest challenges regarding security in the IIoT. An important challenge is the reliable and smooth IDS-based detection mechanism in the IIoT systems. The building previously recorded IIoT devices' network traffic cannot ensure prevention against attacks in case of cold start or fresh deployment. The literature discusses two primary solutions for preventing cyberattacks: IDS and IPS. Various intrusion detection models have been assessed as part of the review work that has been conducted, which analyses the IDS. We briefly review research challenges and open Blockchain and FL-enabled IDS issues in IIoT systems.

6.2. Establishment of a trusted federation

Generally, federated systems and networks are established under the supervisory role of a centralized coordinator to manage the federation and its functions in a fairway. In other words, the centralized coordinator work as a network fabric that interconnects all participating nodes in the federation. However, the centralized network fabric has intrinsic issues such as single point failure, the privacy of the federation participants' data available in the centralized coordinator, bias of the centralized coordinator in conflict resolution among the federation participants, and transparency of billing/incentives of the federation participants.

Blockchain with the smart contract is a panacea to all the issues discussed owing to the fact that Blockchain possesses characteristics like consensus, provenance, ownership, immutability, finality, and access control of distributed ledgers. How to establish a trusted federation that employs Blockchain as a network fabric to establish, interconnect and sustain distributed intrusion detection federation systems.

Volume of training data owned by participating intrusion detection servers in the established federation.

In the best interest to better serve their customers and generate revenue, security service providers are keen to gather and analyze the vast amounts of network data generated by their users' devices and networks. However, there is a high cost in terms of central server processing and storage for all of this collected data. Evidently, sharing and

collecting user data is limited due to the implementation of administrative and regulatory legislation, including the European General Data Protection Regulation (GDPR), China's Cyber Security Law, and the United States California Consumer Privacy Act. Due to the lack of available training data, the performance of machine learning tasks for network traffic categorization may be negatively impacted. In addition, the commercial competition between different providers of security services is another factor that inhibits data sharing since it is also a fact that the number of samples used for training would be insufficient to produce a generalized model with these limitations in place.

6.3. Diversity in training data and models at industrial IoT

The diversity results from taking into account how various machine learning models that imply variety in the context of learning, activities like classification, and regression that are subject to close supervision are the primary focus. Contrarily, unsupervised activities include multi-view learning tasks, online learning tasks, active learning tasks, reinforcement learning tasks, and semi-supervised or unsupervised tasks. As a result, various learning activities necessitate distinct client/data owner settings. Generally, another heterogeneity affects machine learning tasks, i.e., the data distribution of learning tasks. Due to the dynamics of the federated IDS environment, the training data from varying industrial units and devices intrinsically exhibit the notion of statistical heterogeneity, i.e., non-IID distributions. Nonetheless, there are a number of ways in which this data might be non-IID. One example is skewness in the distribution of labels or features. Statistical heterogeneity in data requires special attention in designing AI solutions to maintain acceptable performances.

6.4. Model heterogeneity

In traditional federated learning, the success of training the global machine learning model relies on participating mobile devices reaching a consensus on a specific model. This enables the aggregation of model weights obtained from mobile applications, contributing to effective training. However, in practice, mobile devices and applications often customize their model architecture to suit their unique learning objectives and application environments. Due to privacy concerns, mobile applications are hesitant to share the intricate details of their model architectures. Consequently, the model architectures from various IDS applications exhibit diverse shapes and structures, posing challenges for conventional federated learning's straightforward aggregation approach. The issue of model heterogeneity in distributed IDS has not received significant attention so far. Nevertheless, based on our experience, we believe that investigating its practical implications is essential for the development of intelligent federated IDS systems.

On the other hand, in a federated IDS setup with multiple IDS, each IDS, as illustrated in Fig. 13 will be using a different type of machine learning algorithm for network intrusion detection. Moreover, it is possible that each IDS may have its own preference for training its predictive model using a distinct machine-learning algorithm. In this case, it is essential to adopt an ontology-driven autonomous interaction approach between agents concerning intrusion detection and the machine learning algorithm selection.

6.5. Ontologies for autonomous interaction of IDS agents

The IDS agents should utilize the ontological representation specified in ML-Schema [31] to represent and interchange information on the ML/FL algorithms, datasets, and experiments that will be followed in the IDS workflow. ML Schema is shown in Fig. 14 represents the machine learning algorithms, the machine learning tasks they address, their implementations and executions, and inputs (e.g., data) and outputs (e.g., models) they specify. Representation of machine learning models through ontological schema enables the efficient discovery and

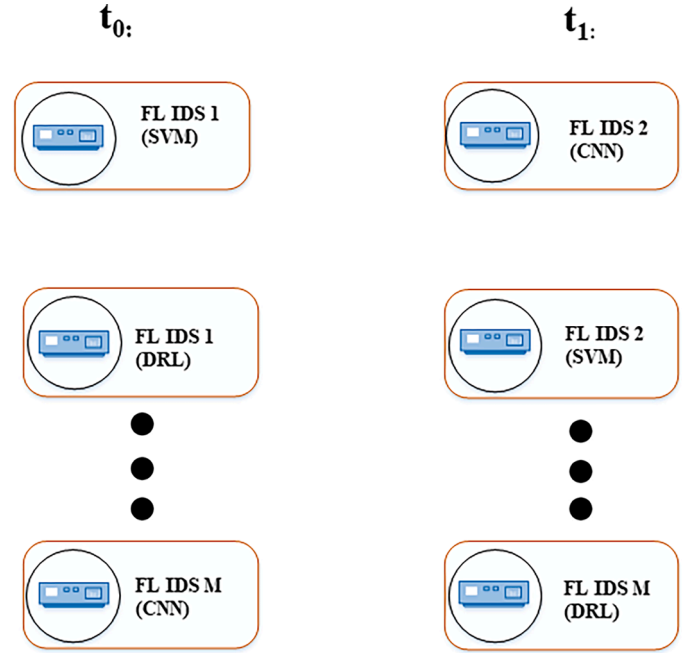


Fig. 13. Model heterogeneity inherent in IDS.

searching of potential participants of interest with the relevant dataset and features for intrusion detection in the federation of intrusion detection systems governed through Blockchain. The ML-Schema published by IDS in the federation is used as the characteristics for match-making and making trade decisions by the human seller agents behind the workers' mobile devices.

6.6. Dynamic reconfiguration

Lack of adaptivity in automated systems has been a serious software industry problem over the years. Typical industrial applications such as IDS consist of many lines of code, which are difficult to standardize the interfaces of IDS with other IDS servers in the federation. When a new IDS system is started and IDS software is developed from scratch, knowledge about the data structures and control flow is sufficient to build the working program. However, in-depth knowledge about the IDS software may fade away rapidly during the IDS program's lifetime. Particularly, the demand for IDS software changes may be high in a changing environment. To fulfill the demand, developers often face a difficult (if not impossible) task: to change incomprehensible lines of code to maintain the IDS in the federations.

7. Conclusion

Integration of the internet of things (IoT) in the industrial internet of things (IIoTs) has paved a tremendous way in the development and advancement of Industry 4.0. However, cybersecurity threats have emerged as a significant stumbling block to the smooth implementation of the industrial revolution brought on by the expanded and quick expansion of the smart industry throughout the world. A solid and holistic system like Intrusion Detection & Intrusion Prevention and a traditional firewall system need an hour to detect malicious and cyberattacks in IIoT systems. We studied the new paradigm of integrated Blockchain and Federated Learning-enabled approaches for Intrusion Detection in IIoT Networks. We analyzed the IDS approaches, highlighting the diversity and complexity of approaches employed to tackle security threats in IoT and IIoT environments. Furthermore, we also presented a discussion on the possible attacks on IIoT systems and recommendations for implementing Blockchain and FL-based IDS systems.

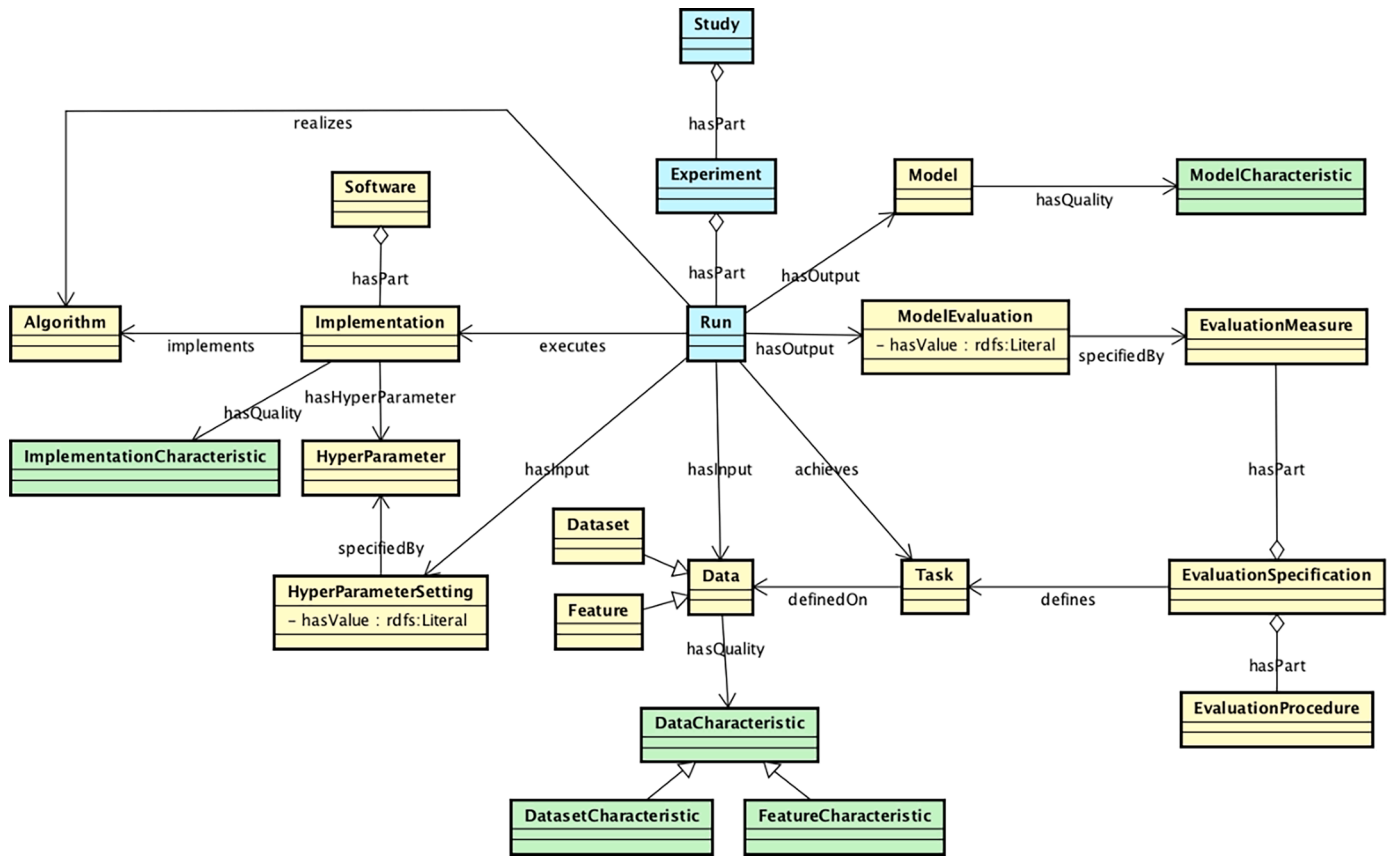


Fig. 14. ML Schema [31].

In the future, we plan to focus on a better cyberattack security solution considering the recommendation and challenges outlined above for the IIoT IDS systems to develop algorithms and methodologies to prevent attacks by using renewed datasets, i.e., NSLKDD, ToN_IoT, UNSWD, and X-IIoTID.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare there is no conflict of interest whatsoever with any party. Furthermore, our respective institutes are aware about the submission of this work to the Journal of AdHoc Networks. Lastly, this research does not involve any human subject.

Data availability

No data was used for the research described in the article.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.adhoc.2023.103320](https://doi.org/10.1016/j.adhoc.2023.103320).

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